

Chapter 3

Newton's Methods

3.1 The Steepest Descent Method

3.1.1 The Steepest Descent Method

The steepest descent method is one of the simplest and the most fundamental minimization methods for unconstrained optimization. Since it uses the negative gradient as its descent direction, it is also called the gradient method.

Suppose that $f(x)$ is continuously differentiable near x_k , and the gradient $g_k \stackrel{\text{def}}{=} \nabla f(x_k) \neq 0$. From the Taylor expansion

$$f(x) = f(x_k) + (x - x_k)^T g_k + o(\|x - x_k\|), \quad (3.1.1)$$

we know that, if we write $x - x_k = \alpha d_k$, then the direction d_k satisfying $d_k^T g_k < 0$ is called a descent direction that is such that $f(x) < f(x_k)$. Fixing α , it follows that the smaller the value $d_k^T g_k$ (i.e., the larger the value $|d_k^T g_k|$) is, the faster the function value decreases. By the Cauchy-Schwartz inequality

$$|d_k^T g_k| \leq \|d_k\| \|g_k\|, \quad (3.1.2)$$

we have that the value $d_k^T g_k$ is the smallest if and only if $d_k = -g_k$. Therefore $-g_k$ is the steepest descent direction.

The iterative scheme of the steepest descent method is

$$x_{k+1} = x_k - \alpha_k g_k. \quad (3.1.3)$$

In the following we give the algorithm.

Algorithm 3.1.1 (*The Steepest Descent Method*)

Step 0. Let $0 < \varepsilon \ll 1$ be the termination tolerance. Given an initial point $x_0 \in R^n$. Set $k = 0$.

Step 1. If $\|g_k\| \leq \varepsilon$, stop ; otherwise let $d_k = -g_k$.

Step 2. Find the steplength factor α_k , such that

$$f(x_k + \alpha_k d_k) = \min_{\alpha \geq 0} f(x_k + \alpha d_k);$$

Step 3. Compute $x_{k+1} = x_k + \alpha_k d_k$.

Step 4. $k := k + 1$, return to Step 1. $\square$

3.1.2 Convergence of the Steepest Descent Method

The steepest descent method is of importance in the area of optimization from the theoretical point of view. The importance of its convergence theory is not only in itself but also in other optimization methods. In the following, we discuss the global convergence and local convergence rate of the steepest descent method.

Theorem 3.1.2 (*Global convergence theorem of the steepest descent method*)
Let $f \in C^1$. Then each accumulation point of the iterative sequence $\{x_k\}$ generated by the steepest descent Algorithm 3.1.1 with exact line search is a stationary point.

Proof. Let $\bar{x}$ be any accumulation point of $\{x_k\}$ and K an infinite index set such that $\lim_{k \in K} x_k = \bar{x}$. Set $d_k = -\nabla f(x_k)$. Since $f \in C^1$, the sequence $\{d_k \mid k \in K\}$ is uniformly bounded and $\|d_k\| = \|\nabla f(x_k)\|$. Since the assumptions of Theorem 2.2.3 are satisfied, it follows that $\|\nabla f(\bar{x})\|^2 = 0$, i.e., $\nabla f(\bar{x}) = 0$. $\square$

Theorem 3.1.3 (*Global convergence theorem of the steepest descent method*)
Let $f(x)$ be twice continuously differentiable in R^n and $\|\nabla^2 f(x)\| \leq M$ for a positive constant M . Given any initial x_0 and $\varepsilon > 0$. Then the sequence generated from Algorithm 3.1.1 terminates in finitely many iterations, or $\lim_{k \rightarrow \infty} f(x_k) = -\infty$, or $\lim_{k \rightarrow \infty} \nabla f(x_k) = 0$.

Proof. Consider the infinite case. From Algorithm 3.1.1 and Theorem 2.2.2, we have

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2M} \|\nabla f(x_k)\|^2.$$

Then

$$f(x_0) - f(x_k) = \sum_{i=0}^{k-1} [f(x_i) - f(x_{i+1})] \geq \frac{1}{2M} \sum_{i=0}^{k-1} \|\nabla f(x_i)\|^2.$$

Taking limits yields either $\lim_{k \rightarrow \infty} f(x_k) = -\infty$ or $\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0$. The result then follows. $\square$

Instead of the exact line search in Step 2 of Algorithm 3.1.1, the steepest descent method can also use inexact line search technique. For this case, the global convergence is given below.

Theorem 3.1.4 (*Convergence theorem of the steepest descent method with inexact line search*)

Let $f \in C^1$. Consider the steepest descent method with inexact line search. Then each accumulation point of the sequence $\{x_k\}$ is a stationary point.

Proof. It follows directly from Theorem 2.5.4. $\square$

Unfortunately, the global convergence does not guarantee that the steepest descent method is an effective method. We can clearly find this problem from the following analysis and the local convergence rate theorem.

In fact, the steepest descent direction is only a local property of the algorithm. For many problems, the steepest descent method is not the actual “steepest”, but is very slow. Although the method usually works well in the early steps, as a stationary point is approached, it descends very slowly with zigzagging phenomena. This zigzagging phenomena is illustrated in Figure 3.1.1 for the problem

$$\min (x_1 - 2)^4 + (x_1 - 2x_2)^2,$$

in which zigzagging occurs along the valley shown by the dotted lines.

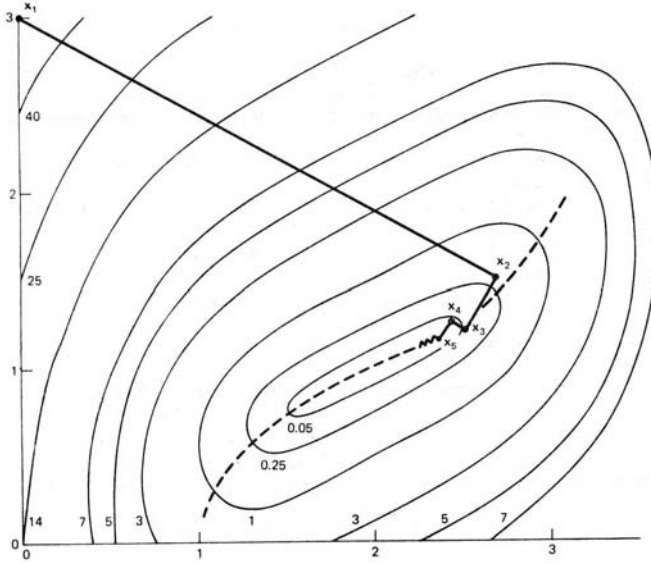


Figure 3.1.1 Zigzagging in the steepest descent method

In fact, the zigzagging of the steepest descent method can be explained by the following facts. Since, from exact line search, one has

$$g_{k+1}^T d_k = 0,$$

then

$$g_{k+1}^T g_k = d_{k+1}^T d_k = 0. \quad (3.1.4)$$

This shows that two gradients are orthogonal to each other on the successive iterates, and thus two successive directions are also orthogonal, which leads to the zigzagging. When the stationary point is approached, $\|g_k\|$ will be very small. By means of the expression

$$f(x_k + \alpha d) = f(x_k) + \alpha g_k^T d + o(\|\alpha d\|), \quad (3.1.5)$$

it is easy to see that the first order term $\alpha g_k^T d = -\alpha \|g_k\|^2$ is of a very small order of magnitude. Hence the descent of f is very small.

Next, we discuss the convergence rate of the steepest descent method, first for the case of a quadratic function and then for the case of a general function.

When the objective function is quadratic, the convergence rate of the steepest descent method depends on the ratio of the longest axis and the

shortest axis of the ellipsoid which corresponds to the contour of the objective function. The bigger the ratio is, the slower the descent is. The following theorem indicates this fact and says that the steepest descent method converges linearly.

Theorem 3.1.5 *(The convergence rate theorem of the steepest descent method for the case of a quadratic function)*

Consider the unconstrained minimization problem

$$\min_{x \in R^n} f(x) = \frac{1}{2}x^T Gx, \quad (3.1.6)$$

where G is an $n \times n$ symmetric and positive definite matrix. Let λ_1 and λ_n be the largest and the smallest eigenvalues of G respectively. Let x^* be the solution of the problem (3.1.6). Then the sequence $\{x_k\}$ generated by the steepest descent method converges to x^* , the convergence rate is at least linear, and the following bounds hold:

$$\frac{f(x_{k+1}) - f(x^*)}{f(x_k) - f(x^*)} \leq \frac{(\kappa - 1)^2}{(\kappa + 1)^2} = \frac{(\lambda_1 - \lambda_n)^2}{(\lambda_1 + \lambda_n)^2}, \quad (3.1.7)$$

$$\frac{\|x_{k+1} - x^*\|_G}{\|x_k - x^*\|_G} \leq \frac{\kappa - 1}{\kappa + 1} = \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right), \quad (3.1.8)$$

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq \sqrt{\kappa} \frac{\kappa - 1}{\kappa + 1} = \sqrt{\frac{\lambda_1}{\lambda_n}} \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right), \quad (3.1.9)$$

where $\kappa = \lambda_1/\lambda_n$.

Proof. Consider the minimization of (3.1.6); we have

$$x_{k+1} = x_k - \alpha_k g_k, \quad (3.1.10)$$

with

$$\alpha_k = \frac{g_k^T g_k}{g_k^T G g_k} \quad (3.1.11)$$

and $g_k = Gx_k$.

$$\begin{aligned} \frac{f(x_k) - f(x_{k+1})}{f(x_k)} &= \frac{\frac{1}{2}x_k^T Gx_k - \frac{1}{2}(x_k - \alpha_k g_k)^T G(x_k - \alpha_k g_k)}{\frac{1}{2}x_k^T Gx_k} \\ &= \frac{\alpha_k g_k^T Gx_k - \frac{1}{2}\alpha_k^2 g_k^T Gg_k}{\frac{1}{2}x_k^T Gx_k} \end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{(g_k^T g_k)^2}{g_k^T G g_k} - \frac{1}{2} \frac{(g_k^T g_k)^2}{g_k^T G g_k}}{\frac{1}{2} g_k^T G^{-1} g_k} \\
&= \frac{(g_k^T g_k)^2}{(g_k^T G g_k)(g_k^T G^{-1} g_k)}. \tag{3.1.12}
\end{aligned}$$

By using Kantorovich inequality (3.1.33), we have immediately that

$$\frac{f(x_{k+1})}{f(x_k)} = \left[1 - \frac{(g_k^T g_k)^2}{(g_k^T G g_k)(g_k^T G^{-1} g_k)} \right] \tag{3.1.13}$$

$$\leq \left[1 - \frac{4\lambda_1\lambda_n}{(\lambda_1 + \lambda_n)^2} \right] = \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right)^2, \tag{3.1.14}$$

which is just (3.1.7).

By using (3.1.13), it is not difficult to get (3.1.8) and (3.1.9). In fact, let $e_k = x_k - x^*$, $\forall k \geq 0$. Noting that G is symmetric and positive definite, we have

$$\lambda_n e_k^T e_k \leq e_k^T G e_k \leq \lambda_1 e_k^T e_k. \tag{3.1.15}$$

Since $x^* = 0$, we have

$$\|x_k - x^*\|_G^2 = e_k^T G e_k = x_k^T G x_k = 2f(x_k). \tag{3.1.16}$$

So, it follows from (3.1.15) that

$$\lambda_n \|x_k - x^*\|^2 \leq 2f(x_k) \leq \lambda_1 \|x_k - x^*\|^2, \quad \forall k \geq 0. \tag{3.1.17}$$

From (3.1.13), (3.1.16) and (3.1.17), we get

$$\frac{\lambda_n \|x_{k+1} - x^*\|^2}{\lambda_1 \|x_k - x^*\|^2} \leq \frac{\|x_{k+1} - x^*\|_G^2}{\|x_k - x^*\|_G^2} \leq \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right)^2, \tag{3.1.18}$$

which gives (3.1.8) and (3.1.9). $\square$

If we consider, more generally, the objective function with the form

$$f(x) = \frac{1}{2} x^T G x - b^T x, \tag{3.1.19}$$

where G is an $n \times n$ symmetric positive definite matrix and $b \in R^n$, the above theorem is also true.

If the objective function is generalized to the non-quadratic case, we also can get the linear convergence rate of the steepest descent method.

Theorem 3.1.6 *Let $f(x)$ satisfy the assumptions of Theorem 2.2.8. If the sequence $\{x_k\}$ generated from the steepest descent method converges to x^* , then the convergence rate is at least linear.*

Proof. It is a direct result from Theorem 2.2.8. $\square$

The above convergence rate theorem of the steepest descent method for a general function can also be described as follows.

Theorem 3.1.7 *Let $f(x)$ be twice continuously differentiable near x^* with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ positive definite. Let the sequence $\{x_k\}$ generated by the steepest descent method converge to x^* . Let*

$$\frac{f(x_{k+1}) - f(x^*)}{f(x_k) - f(x^*)} = \beta_k. \quad (3.1.20)$$

Then $\beta_k < 1, \forall k$ and

$$\limsup_{k \rightarrow +\infty} \beta_k \leq \frac{M - m}{M} < 1, \quad (3.1.21)$$

where M and m satisfy

$$0 < m \leq \lambda_n \leq \lambda_1 \leq M, \quad (3.1.22)$$

and λ_n and λ_1 are respectively the smallest and the largest eigenvalues of $\nabla^2 f(x)$.

Proof. From Theorem 2.2.2, we have

$$\begin{aligned} [f(x_k) - f(x^*)] - [f(x_{k+1}) - f(x^*)] &= f(x_k) - f(x_{k+1}) \\ &\geq \frac{1}{2M} \|\nabla f(x_k)\|^2, \end{aligned} \quad (3.1.23)$$

which is, by the definition of β_k , that

$$(1 - \beta_k)[f(x_k) - f(x^*)] \geq \frac{1}{2M} \|\nabla f(x_k)\|^2.$$

Hence, by the assumption of f , we get

$$\beta_k \leq 1 - \frac{\|\nabla f(x_k)\|^2}{2M[f(x_k) - f(x^*)]} < 1. \quad (3.1.24)$$

Now suppose that $(x_k - x^*)/\|x_k - x^*\| \rightarrow \bar{d}$. It is obvious that

$$\|\nabla f(x_k)\|^2 = \|x_k - x^*\|^2 (\|\nabla^2 f(x^*)\bar{d}\|^2 + o(1))$$

and

$$f(x_k) - f(x^*) = \frac{1}{2}\|x_k - x^*\|^2 (\bar{d}^T \nabla^2 f(x^*) \bar{d} + o(1)).$$

Using the above equalities and (3.1.22) yields

$$\lim_{k \rightarrow \infty} \frac{\|\nabla f(x_k)\|^2}{f(x_k) - f(x^*)} = \frac{2\|\nabla^2 f(x^*)\bar{d}\|^2}{\bar{d}^T \nabla^2 f(x^*) \bar{d}} \geq 2m. \quad (3.1.25)$$

Hence, it follows from (3.1.24) and (3.1.25) that

$$\begin{aligned} \limsup_{k \rightarrow \infty} \beta_k &\leq 1 - \liminf_{k \rightarrow \infty} \frac{\|\nabla f(x_k)\|^2}{2M[f(x_k) - f(x^*)]} \\ &\leq 1 - \frac{m}{M} < 1. \end{aligned}$$

We complete the proof. $\square$

3.1.3 Barzilai and Borwein Gradient Method

From the above discussions we know that the classical steepest descent method performs poorly, converges linearly, and is badly affected by ill-conditioning.

Barzilai and Borwein [8] presented a two-point step size gradient method, which is called usually the Barzilai-Borwein (or BB) gradient method. In the method, the step size is derived from a two-point approximation to the secant equation underlying quasi-Newton methods (see Chapter 5).

Consider the gradient iteration form

$$x_{k+1} = x_k - \alpha_k g_k \quad (3.1.26)$$

which can be written as

$$x_{k+1} = x_k - D_k g_k, \quad (3.1.27)$$

where $D_k = \alpha_k I$. In order to make the matrix D_k have quasi-Newton property, we compute α_k such that

$$\min \|s_{k-1} - D_k y_{k-1}\|. \quad (3.1.28)$$

This yields that

$$\alpha_k = \frac{s_{k-1}^T y_{k-1}}{y_{k-1}^T y_{k-1}}, \quad (3.1.29)$$

where $s_{k-1} = x_k - x_{k-1}$, $y_{k-1} = g_k - g_{k-1}$.

By symmetry, we may minimize $\|D_k^{-1}s_{k-1} - y_{k-1}\|$ with respect to α_k and get

$$\alpha_k = \frac{s_{k-1}^T s_{k-1}}{s_{k-1}^T y_{k-1}}. \quad (3.1.30)$$

The above description produces the following algorithm.

Algorithm 3.1.8 (*The Barzilai-Borwein gradient method*)

Step 0. Given $x_0 \in R^n$, $0 < \varepsilon \ll 1$. Set $k = 0$.

Step 1. If $\|g_k\| \leq \varepsilon$, stop ; otherwise let $d_k = -g_k$.

Step 2. If $k = 0$, find α_0 by line search; otherwise compute α_k by (3.1.29) or (3.1.30).

Step 3. Set $x_{k+1} = x_k + \alpha_k d_k$.

Step 4. $k := k + 1$, return to Step 1. $\square$

It is easy to see that in this method no matrix computations and no line searches (except $k = 0$) are required. The Barzilai-Borwein method is, in fact, a gradient method, but requires less computational work, and greatly speeds up the convergence of the gradient method. Barzilai and Borwein [8] proved that the above algorithm is R -superlinearly convergent for the quadratic case.

In the general non-quadratic case, a globalization strategy based on non-monotone line search is suitable to Barzilai-Borwein gradient method. In addition, in general non-quadratic case, α_k computed by (3.1.29) or (3.1.30) can be unacceptably large or small. Therefore, we must assume that α_k satisfies the condition

$$0 < \alpha^{(l)} \leq \alpha_k \leq \alpha^{(u)}, \quad \text{for all } k,$$

where $\alpha^{(l)}$ and $\alpha^{(u)}$ are previously determined numbers.

If we employ the iteration

$$x_{k+1} = x_k - \frac{1}{\alpha_k} g_k = x_k - \lambda_k g_k \quad (3.1.31)$$

with

$$\alpha_k = \frac{s_{k-1}^T y_{k-1}}{s_{k-1}^T s_{k-1}}, \quad \lambda_k = \frac{1}{\alpha_k}, \quad (3.1.32)$$

note that $s_k = -\frac{1}{\alpha_k} g_k = -\lambda_k g_k$, then we have

$$\alpha_{k+1} = \frac{s_k^T y_k}{s_k^T s_k} = \frac{-\lambda_k g_k^T y_k}{\lambda_k^2 g_k^T g_k} = -\frac{g_k^T y_k}{\lambda_k g_k^T g_k}.$$

Now we give the following Barzilai-Borwein gradient algorithm with non-monotone globalization.

Algorithm 3.1.9 (*The Barzilai-Borwein gradient algorithm with nonmonotone linesearch*)

Step 0. Given $x_0 \in R^n$, $0 < \varepsilon \ll 1$, an integer $M \geq 0$, $\rho \in (0, 1)$, $\delta > 0$, $0 < \sigma_1 < \sigma_2 < 1$, $\alpha^{(l)}, \alpha^{(u)}$. Set $k = 0$.

Step 1. If $\|g_k\| \leq \varepsilon$, stop.

Step 2. If $\alpha_k \leq \alpha^{(l)}$ or $\alpha_k \geq \alpha^{(u)}$ then set $\alpha_k = \delta$.

Step 3. Set $\lambda = 1/\alpha_k$.

Step 4. (nonmonotone line search) If

$$f(x_k - \lambda g_k) \leq \max_{0 \leq j \leq \min(k, M)} f(x_{k-j}) - \rho \lambda g_k^T g_k,$$

then set

$$\lambda_k = \lambda, \quad x_{k+1} = x_k - \lambda_k g_k,$$

and go to Step 6.

Step 5. Choose $\sigma \in [\sigma_1, \sigma_2]$, set $\lambda = \sigma \lambda$, and go to Step 4.

Step 6. Set $\alpha_{k+1} = -(g_k^T y_k)/(\lambda_k g_k^T g_k)$, $k := k + 1$, return to Step 1.

□

Obviously, the above algorithm is globally convergent.

3.1.4 Appendix: Kantorovich Inequality

We conclude this section with a famous Kantorovich Inequality which is used in the proof of Theorem 3.1.5.

Theorem 3.1.10 (*Kantorovich Inequality*) *Let G be an $n \times n$ symmetric positive definite matrix with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. Then, for any $x \in \mathbb{R}^n$, the following inequality holds:*

$$\frac{(x^T x)^2}{(x^T G x)(x^T G^{-1} x)} \geq \frac{4\lambda_1 \lambda_n}{(\lambda_1 + \lambda_n)^2}. \quad (3.1.33)$$

Proof. Let the spectral decomposition of G be

$$G = U \Lambda U.$$

Set $x = Uy$, then

$$\begin{aligned} \frac{(x^T x)^2}{(x^T G x)(x^T G^{-1} x)} &= \frac{(y^T y)^2}{(y^T \Lambda y)(y^T \Lambda^{-1} y)} \\ &= \frac{(\sum_{i=1}^n y_i^2)^2}{(\sum_{i=1}^n \lambda_i y_i^2)(\sum_{i=1}^n y_i^2 / \lambda_i)}. \end{aligned} \quad (3.1.34)$$

Let

$$\xi_i = \frac{y_i^2}{\sum_{i=1}^n y_i^2}, \quad \phi(\lambda) = \frac{1}{\lambda}, \quad (3.1.35)$$

then (3.1.34) becomes

$$\frac{(x^T x)^2}{(x^T G x)(x^T G^{-1} x)} = \frac{1}{(\sum_{i=1}^n \lambda_i \xi_i)(\sum_{i=1}^n \phi(\lambda_i) \xi_i)}. \quad (3.1.36)$$

Below we use the convexity of ϕ to estimate the lower bound of the right-hand side of (3.1.36). Let

$$\lambda = \sum_{i=1}^n \lambda_i \xi_i, \quad \lambda_\phi = \sum_{i=1}^n \phi(\lambda_i) \xi_i. \quad (3.1.37)$$

Since $\xi_i \geq 0$ ($i = 1, \dots, n$) and $\sum_{i=1}^n \xi_i = 1$, we have $\lambda_n \leq \lambda \leq \lambda_1$. Then each λ_i can be represented as a convex combination of λ_1 and λ_n :

$$\lambda_i = \frac{\lambda_1 - \lambda_i}{\lambda_1 - \lambda_n} \lambda_n + \frac{\lambda_i - \lambda_n}{\lambda_1 - \lambda_n} \lambda_1.$$

From the convexity of ϕ , we have obviously

$$\phi(\lambda_i) \leq \frac{\lambda_1 - \lambda_i}{\lambda_1 - \lambda_n} \phi(\lambda_n) + \frac{\lambda_i - \lambda_n}{\lambda_1 - \lambda_n} \phi(\lambda_1). \quad (3.1.38)$$

Then, it follows from (3.1.37), (3.1.38) and (3.1.35) that

$$\begin{aligned} \lambda_\phi &\leq \sum_{i=1}^n \left[\frac{\lambda_1 - \lambda_i}{\lambda_1 - \lambda_n} \phi(\lambda_n) + \frac{\lambda_i - \lambda_n}{\lambda_1 - \lambda_n} \phi(\lambda_1) \right] \xi_i \\ &= \sum_{i=1}^n \frac{\lambda_1 + \lambda_n - \lambda_i}{\lambda_1 \lambda_n} \xi_i \\ &= \frac{\lambda_1 + \lambda_n - \lambda}{\lambda_1 \lambda_n}. \end{aligned} \quad (3.1.39)$$

Therefore, by (3.1.36), (3.1.37) and (3.1.39) we obtain

$$\begin{aligned} \frac{(x^T x)^2}{(x^T G x)(x^T G^{-1} x)} &= \frac{1}{\lambda \lambda_\phi} \geq \frac{\lambda_1 \lambda_n}{\lambda(\lambda_1 + \lambda_n - \lambda)} \\ &\geq \frac{\lambda_1 \lambda_n}{\max_{\lambda \in [\lambda_n, \lambda_1]} \lambda(\lambda_1 + \lambda_n - \lambda)} = \frac{4\lambda_1 \lambda_n}{(\lambda_1 + \lambda_n)^2}, \end{aligned}$$

which is our result. $\square$

3.2 Newton's Method

The basic idea of Newton's method for unconstrained optimization is to iteratively use the quadratic approximation $q^{(k)}$ to the objective function f at the current iterate x_k and to minimize the approximation $q^{(k)}$.

Let $f : R^n \rightarrow R$ be twice continuously differentiable, $x_k \in R^n$, and the Hessian $\nabla^2 f(x_k)$ positive definite. We model f at the current point x_k by the quadratic approximation $q^{(k)}$,

$$f(x_k + s) \approx q^{(k)}(s) = f(x_k) + \nabla f(x_k)^T s + \frac{1}{2} s^T \nabla^2 f(x_k) s, \quad (3.2.1)$$

where $s = x - x_k$. Minimizing $q^{(k)}(s)$ yields

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \quad (3.2.2)$$

which is Newton's formula. Set

$$G_k = \nabla^2 f(x_k), \quad g_k = \nabla f(x_k). \quad (3.2.3)$$

Then we write (3.2.2) as

$$x_{k+1} = x_k - G_k^{-1} g_k, \quad (3.2.4)$$

where $s_k = x_{k+1} - x_k = -G_k^{-1} g_k$ is a Newton's direction. Clearly, the Newton's direction is a descent direction because it satisfies $g_k^T s_k = -g_k^T G_k^{-1} g_k < 0$ if G_k is positive definite. Please note, in the remainder of this book, the first and the second derivatives of f will be denoted by

$$g(x) \triangleq \nabla f(x), \quad G(x) \triangleq \nabla^2 f(x) \quad (3.2.5)$$

for convenience, if they exist.

The corresponding algorithm is stated as follows.

Algorithm 3.2.1 (*Newton's Method*)

Step 1. Given $x_0 \in R^n, \epsilon > 0, k := 0$;

Step 2. If $\|g_k\| \leq \epsilon$, stop;

Step 3. Solve $G_k s = -g_k$ for s_k ;

Step 4. Set $x_{k+1} = x_k + s_k$;

Step 5. $k := k + 1$, go to Step 2. $\square$

Obviously, Newton's method can be regarded as a steepest descent method under the ellipsoid norm $\|\cdot\|_{G_k}$. In fact, for $f(x_k + s) \approx f(x_k) + g_k^T s$, we regard s_k as the solution of the minimization problem

$$\min_{s \in R^n} \frac{g_k^T s}{\|s\|}. \quad (3.2.6)$$

The solution of (3.2.6) depends on the norm. If we employ l_2 norm, then we get $s_k = -g_k$ and the resultant method is the steepest descent method. If we employ the ellipsoid norm $\|\cdot\|_{G_k}$, then we get $s_k = -G_k^{-1} g_k$ which is just the Newton's method. In fact, in this case, (3.2.6) is equivalent to

$$\begin{aligned} \min_{s \in R^n} \quad & g_k^T s \\ \text{s.t.} \quad & \|s\|_{G_k} \leq 1. \end{aligned}$$

Note that, by (1.2.36), we have that

$$(g_k^T s)^2 \leq (g_k^T G_k^{-1} g_k)(s^T G_k s)$$

and that $g_k^T s$ will be the smallest when $s = -G_k^{-1} g_k$. The above discussion gives us a clear explanation.

For the positive definite quadratic function, Newton's method can reach the minimizer with one iteration. However, for a general non-quadratic function, it is not sure that Newton's method can reach the minimizer with finite iterations. Fortunately, since the objective function is approximate to a quadratic function near the minimizer, then if the starting point is close to the minimizer the Newton's method will converge rapidly. The following theorem shows the local convergence and the quadratic convergence rate of Newton's method.

Theorem 3.2.2 (*Convergence Theorem of Newton's Method*) *Let $f \in C^2$ and x_k be close enough to the solution x^* of the minimization problem with $g(x^*) = 0$. If the Hessian $G(x^*)$ is positive definite and $G(x)$ satisfies Lipschitz condition*

$$|G_{ij}(x) - G_{ij}(y)| \leq \beta \|x - y\|, \text{ for some } \beta, \text{ for all } i, j \quad (3.2.7)$$

where $G_{ij}(x)$ is the (i, j) -element of $G(x)$, then for all k , Newton's iteration (3.2.4) is well-defined; the generated sequence $\{x_k\}$ converges to x^* with a quadratic rate.

Proof. Let $h_k = x_k - x^*$. From Taylor's formula, it follows that

$$0 = g(x^*) = g_k - G_k h_k + O(\|h_k\|^2).$$

Since $f \in C^2$, x_k is close enough to x^* , and $G(x^*)$ is positive definite, it is reasonable to assume that x_k is in the neighborhood of x^* , G_k positive definite, G_k^{-1} upper bounded. Hence the k -th Newton's iteration exists. Multiplying through by G_k^{-1} yields

$$\begin{aligned} 0 &= G_k^{-1} g_k - h_k + O(\|h_k\|^2) \\ &= -s_k - h_k + O(\|h_k\|^2) \\ &= -h_{k+1} + O(\|h_k\|^2). \end{aligned}$$

By definition of $O(\cdot)$, there is a constant C such that

$$\|h_{k+1}\| \leq C \|h_k\|^2. \quad (3.2.8)$$

If $x_k \in \Omega = \{x \mid \|h\| \leq \gamma/C, h = x - x^*, \gamma \in (0, 1)\}$, then

$$\|h_{k+1}\| \leq \gamma \|h_k\| \leq \gamma^2/C < \gamma/C. \quad (3.2.9)$$

Hence $x_{k+1} \in \Omega$. By induction on k , Newton's iteration is well-defined for all k , and $\|h_k\| \rightarrow 0$ as $k \rightarrow \infty$. Therefore the iteration converges. Also, (3.2.8) shows that the convergence rate of the iteration sequence is quadratic. $\square$

Note that Newton's method is a local method. When the starting point is far away from the solution, it is not sure that G_k is positive definite and Newton's direction d_k is a descent direction. Hence the convergence is not guaranteed. Since, as we know, the line search is a global strategy, we can employ Newton's method with line search to guarantee the global convergence. However it should be noted that only when the step size sequence $\{\alpha_k\}$ converges to 1, Newton's method is convergent with the quadratic rate. Newton's iteration with line search is as follows:

$$d_k = -G_k^{-1}g_k, \quad (3.2.10)$$

$$x_{k+1} = x_k + \alpha_k d_k, \quad (3.2.11)$$

where α_k is a step size. The formula (3.2.10)–(3.2.11) corresponds to the following algorithm.

Algorithm 3.2.3 (*Newton's Method with Line Search*)

Step 1. Initial step: given $x_0 \in R^n, \epsilon > 0$, set $k := 0$.

Step 2. Compute g_k . If $\|g_k\| \leq \epsilon$, stop and output x_k ; otherwise go to Step 3.

Step 3. Solve $G_k d = -g_k$ for d_k .

Step 4. Line search step: find α_k such that

$$f(x_k + \alpha_k d_k) = \min_{\alpha \geq 0} f(x_k + \alpha d_k).$$

Step 5. Set $x_{k+1} = x_k + \alpha_k d_k$, $k := k + 1$, go to Step 2. $\square$

Next, we prove the above Algorithm 3.2.3 is globally convergent.

Theorem 3.2.4 *Let $f : R^n \rightarrow R$ be twice continuously differentiable on open convex set $D \subset R^n$. Assume that for any $x_0 \in D$ there exists a constant $m > 0$ such that $f(x)$ satisfies*

$$u^T \nabla^2 f(x) u \geq m \|u\|^2, \quad \forall u \in R^n, x \in L(x_0), \quad (3.2.12)$$

where $L(x_0) = \{x \mid f(x) \leq f(x_0)\}$ is the corresponding level set. Then the sequence $\{x_k\}$ generated by Algorithm 3.2.3 satisfies

1. when $\{x_k\}$ is a finite sequence, $g_k = 0$ for some k ;
2. when $\{x_k\}$ is an infinite sequence, $\{x_k\}$ converges to the unique minimizer x^* of f .

Proof. First, from (3.2.12), we know that $f(x)$ is a strictly convex function on R^n , and hence its stationary point is the unique global minimizer.

Also, from the assumption, it follows that the level set $L(x_0)$ is a bounded closed convex set. Since $\{f(x_k)\}$ is monotonic descent, then $\{x_k\} \subset L(x_0)$ and $\{x_k\}$ is bounded. Therefore there exists a limit point $\bar{x} \in L(x_0)$ with $x_k \rightarrow \bar{x}$, and further $f(x_k) \rightarrow f(\bar{x})$. Also since $f \in C^2(D)$, by Theorem 2.2.4, we have $g_k \rightarrow g(\bar{x}) = 0$. Finally, note that the stationary point is unique, then the whole sequence $\{x_k\}$ converges to $\bar{x}$ which is the unique minimizer. $\square$

Similarly, if we employ inexact line search rule (2.5.3) and (2.5.7), it follows from (2.5.22) that

$$f(x_k) - f(x_k + \alpha_k d_k) \geq \bar{\eta} \|g_k\|^2 \cos^2 \langle d_k, -g_k \rangle, \quad (3.2.13)$$

where $\bar{\eta}$ is some constant independent of k . In this case the global convergence still holds.

Theorem 3.2.5 *Let $f : R^n \rightarrow R$ be twice continuously differentiable on an open convex set $D \subset R^n$. Assume that for any $x_0 \in R^n$, there exists $m > 0$ such that $f(x)$ satisfies (3.2.12) on the level set $L(x_0)$. If the line search employed satisfies (3.2.13), then the sequence $\{x_k\}$ generated from Newton's algorithm satisfies*

$$\lim_{k \rightarrow \infty} \|g_k\| = 0, \quad (3.2.14)$$

and $\{x_k\}$ converges to the unique minimizer of $f(x)$.

Proof. Since $f(x)$ satisfies (3.2.12), we see that $f(x)$ is uniformly convex on $L(x_0)$. Also, from (3.2.13), it follows that $f(x)$ is strictly monotonically descending and further that $\{x_k\}$ is bounded. Therefore there exists a constant $M > 0$ such that

$$\|G_k\| \leq M \quad \forall k. \quad (3.2.15)$$

From (3.2.10), (3.2.12) and (3.2.15), it follows that

$$\begin{aligned} \cos \langle d_k, -g_k \rangle &= \frac{-d_k^T g_k}{\|d_k\| \|g_k\|} = \frac{g_k^T G_k^{-1} g_k}{\|G_k^{-1} g_k\| \|g_k\|} \\ &= \frac{d_k^T G_k d_k}{\|d_k\| \|G_k d_k\|} \geq \frac{m}{M}. \end{aligned} \quad (3.2.16)$$

Hence, by (3.2.13) and (3.2.16), we have

$$\infty > \sum_{k=0}^{\infty} [f(x_k) - f(x_{k+1})] \geq \sum_{k=0}^{\infty} \bar{\eta} \frac{m^2}{M^2} \|g_k\|^2, \quad (3.2.17)$$

which shows (3.2.14). Note that $f(x)$ is uniformly convex, then $f(x)$ has only one stationary point, and (3.2.14) indicates that $\{x_k\}$ converges to the unique minimizer x^* of f . $\square$

3.3 Modified Newton's Method

The main difficulty faced by Newton's method is that the Hessian G_k is not positive definite. In this case, it is not sure that the model function has minimizers. When G_k is indefinite, the model function will be unbounded.

To overcome these difficulties, there are several modified schemes.

Goldstein-Price Method

Goldstein and Price [159] presented a modified method: when G_k is not positive definite, the steepest descent direction $-g_k$ is used. If we combine this strategy with the angle rule

$$\theta \leq \frac{\pi}{2} - \mu, \text{ for some } \mu > 0,$$

where θ is the angle between $-g_k$ and d_k , we can determine the direction d_k as follows:

$$d_k = \begin{cases} -G_k^{-1} g_k, & \text{if } \cos \theta \geq \eta, \\ -g_k, & \text{otherwise,} \end{cases} \quad (3.3.1)$$

where $\eta > 0$ is a given constant. Then the consultant direction d_k satisfies $\cos \theta \geq \eta$ and the angle rule is satisfied, and thus the corresponding algorithm is convergent.

Goldfeld et al. Method

Goldfeld et al. [156] presented another modified Newton's method. Their method does not substitute the steepest descent method for Newton's method, but makes the Newton's direction $-G_k^{-1}g_k$ turn to the steepest descent direction $-g_k$. More precisely, when G_k is not positive definite, one changes the model Hessian G_k to $G_k + \nu_k I$, where $\nu_k > 0$ such that $G_k + \nu_k I$ is positive definite and well-conditioned. Ideally, ν_k is not much larger than the smallest ν that makes $G_k + \nu I$ positive definite and well-conditioned. The framework of the algorithm is as follows.

Algorithm 3.3.1 (Modified Newton's Method)

Initial step: Given an initial point $x_0 \in R^n$.

k-th step:

(1) Set $\bar{G}_k = G_k + \nu_k I$, where

$\nu_k = 0$, if G_k is positive definite;

$\nu_k > 0$, otherwise.

(2) Solve $\bar{G}_k d = -g_k$ for d_k .

(3) Set $x_{k+1} = x_k + d_k$. $\square$

In the above algorithm, the smallest possible ν_k is slightly larger than the magnitude of the most negative eigenvalue of G_k . We suggest applying the Gill-Murray's modified Cholesky factorization to G_k to determine ν_k , which results in

$$G_k + E = LDL^T, \quad (3.3.2)$$

where E is a diagonal matrix with nonnegative diagonal elements (see Gill, Murray and Wright [152]). If $E = 0$, set $\nu_k = 0$; if $E \neq 0$, we can use the Gerschgorin Circle Theorem 1.2.14 to compute an upper bound b_1 of ν_k :

$$b_1 = \left| \min_{1 \leq i \leq n} \left\{ (G_k)_{ii} - \sum_{j \neq i} |(G_k)_{ij}| \right\} \right| \geq \left| \min_i \lambda_i \right|. \quad (3.3.3)$$

In addition, note that

$$b_2 = \max_i \{e_{ii}\} \quad (3.3.4)$$

is also an upper bound of ν_k , where e_{ii} is the i -th diagonal element of E . Then we set

$$\nu_k = \min\{b_1, b_2\}, \quad (3.3.5)$$

and get the positive definite matrix $\bar{G}_k$ and its Cholesky factorization.

In the remainder of this section, we would like to introduce another numerically stable modified Cholesky factorization due to Gill and Murray [149].

It is well-known that the Cholesky factorization $G_k = LDL^T$ of a positive definite matrix G_k can be described as follows:

$$d_{jj} = g_{jj} - \sum_{s=1}^{j-1} d_{ss} l_{js}^2, \quad (3.3.6)$$

$$l_{ij} = \frac{1}{d_{jj}} \left(g_{ij} - \sum_{s=1}^{j-1} d_{ss} l_{js} l_{is} \right), \quad i \geq j+1, \quad (3.3.7)$$

where g_{ij} denote the elements of G_k , d_{jj} the diagonal elements of D . Now we ask the Cholesky factors L and D to satisfy the following two requirements: one is that all the diagonal elements of D are positive; the other is that the elements of the factors are uniformly bounded. That is,

$$d_{kk} > \delta > 0, \quad \forall k \text{ and } |r_{ik}| \leq \beta, \quad i > k, \quad (3.3.8)$$

where $r_{ik} = l_{ik} \sqrt{d_{kk}}$, β is a given positive number and δ is a small positive number.

Below we will describe the j -th step of this factorization. Suppose that the first $j-1$ columns of the factors have been computed, that is, for $k = 1, \dots, j-1$, d_{kk} and l_{ik} ($i = 1, \dots, n$) have been computed and satisfy (3.3.8). Now we compute

$$\gamma_j = |\xi_j - \sum_{s=1}^{j-1} d_{ss} l_{js}^2|, \quad (3.3.9)$$

where ξ_j takes g_{jj} and the test value $\bar{d}$ takes

$$\bar{d} = \max\{\gamma_j, \delta\}. \quad (3.3.10)$$

In order to judge whether to accept $\bar{d}$ as the j -th element of D , we check if $r_{ij} = l_{ij}\sqrt{\bar{d}}$ satisfies (3.3.8). If yes, set $d_{jj} = \bar{d}$ and form the j -th column of L by use of $l_{ij} = r_{ij}/\sqrt{d_{jj}}$; otherwise, set

$$d_{jj} = \left| \xi_j - \sum_{s=1}^{j-1} d_{ss} l_{js}^2 \right|, \quad (3.3.11)$$

where we take $\xi_j = g_{jj} + e_{jj}$ in which e_{jj} is chosen such that $\max |r_{ij}| = \beta$, and also form the j -th column of L as above.

When the above procedure is complete, we obtain a Cholesky factorization of $\bar{G}_k$,

$$\bar{G}_k = LDL^T = G_k + E, \quad (3.3.12)$$

where E is a diagonal matrix with nonnegative diagonal elements e_{jj} . For given G_k , the nonnegative diagonal matrix E depends on the given β . Gill and Murray (1974) prove that if $n > 1$, then

$$\|E(\beta)\|_\infty \leq \left(\frac{\xi}{\beta} + (n-1)\beta \right)^2 + 2(\gamma + (n-1)\beta^2) + \delta, \quad (3.3.13)$$

where ξ and γ are respectively the maximum modules of non-diagonal elements and diagonal elements of G_k . Since, when $\beta^2 = \xi/\sqrt{n^2 - 1}$, the above bound is minimized, then we take β satisfying

$$\beta^2 = \max\{\gamma, \xi/\sqrt{n^2 - 1}, \epsilon_M\} \quad (3.3.14)$$

where ϵ_M denotes the machine precision. Also, note that adding the term ϵ_M in (3.3.14) is to prevent the case in which $\|G_k\|$ is too small.

Now we are in a position to state the modified Cholesky factorization algorithm in which $c_{is} = l_{is}d_{ss}$ ($s = 1, \dots, j; i = j, \dots, n$) are auxiliary variables saved in G_k and we need not increase the storage.

Algorithm 3.3.2 (*Modified Cholesky Factorization due to Gill and Murray (1974)*)

Step 1. Compute β by (3.3.14). Given δ . Set $j := 1, c_{ii} = g_{ii}$ for $i = 1, \dots, n$.

Step 2. Find the smallest index q such that $|c_{qq}| = \max_{j \leq i \leq n} |c_{ii}|$, exchange the q -th and the i -th rows, the q -th and the i -th columns.

Step 3. Compute the j -th row of L and find the maximum module of $l_{ij}d_{jj}$.

Set $l_{js} = c_{js}/d_{ss}, s = 1, \dots, j-1$;

Compute $c_{ij} = g_{ij} - \sum_{s=1}^{j-1} l_{js}c_{is}, i = j+1, \dots, n$;

Set $\theta_j = \max_{j+1 \leq i \leq n} |c_{ij}|$ (if $j = n, \theta_j = 0$).

Step 4. Compute the j -th diagonal element of D :

$$d_{jj} = \max\{\delta, |c_{jj}|, \theta_j^2/\beta^2\};$$

Update the element e_{jj} : $e_{jj} = d_{jj} - c_{jj}$. If $j = n$, stop.

Step 5. Update $c_{ii} = c_{ii} - c_{ij}^2/d_{jj}, i = j+1, \dots, n$;

Set $j := j+1$, go to Step 2. $\square$

The modified Cholesky factorization above needs about $\frac{1}{6}n^3$ arithmetic operations which are almost the same as the normal Cholesky factorization.

Example 3.3.3 *Consider*

$$G_k = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 + 10^{-20} & 3 \\ 2 & 3 & 1 \end{pmatrix}. \quad (3.3.15)$$

By the above Algorithm 3.3.2, we can get $\beta^2 = 1.061$,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0.2652 & 1 & 0 \\ 0.5303 & 0.4295 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3.771 & 0 & 0 \\ 0 & 5.750 & 0 \\ 0 & 0 & 1.121 \end{pmatrix},$$

$$E = \begin{pmatrix} 2.771 & 0 & 0 \\ 0 & 5.016 & 0 \\ 0 & 0 & 2.243 \end{pmatrix}.$$

The difference $\|\bar{G}_k - G_k\|_F = \|E\|_F \approx 6.154$. Since $d_{jj} \geq \delta$ in the modified factorization, it is guaranteed that $\bar{G}_k = G_k + E_k$ is positive definite and the condition number is uniformly bounded, i.e.,

$$\|\bar{G}_k\| \|\bar{G}_k^{-1}\| \leq \kappa, \quad \kappa \geq 0.$$

So, we have

$$-\frac{\nabla f(x_k)^T s_k}{\|s_k\|} \geq \frac{1}{\kappa} \|\nabla f(x_k)\|. \quad (3.3.16)$$

Thus, it follows from the inexact line search, (2.5.19) and (3.3.16) that $\{\nabla f(x_k)\}$ converges to zero.

Theorem 3.3.4 *Let $f : D \subset R^n \rightarrow R$ be twice continuously differentiable on an open set D . Let the level set $\Omega = \{x \mid f(x) \leq f(x_0)\}$ be compact. If the sequence $\{x_k\}$ is generated by the modified Newton's method, then*

$$\lim_{k \rightarrow \infty} \nabla f(x_k) = 0. \quad (3.3.17)$$

3.4 Finite-Difference Newton's Method

The finite-difference Newton's method is to use the finite-difference as an approximation of derivatives in Newton's method.

We first review the finite-difference derivative approximations.

Let $F : R^n \rightarrow R^m$. The (i, j) -component of the Jacobian $J(x)$ of $F(x)$ can be approximated by

$$a_{i,j} = \frac{f_i(x + he_j) - f_i(x)}{h}, \quad (3.4.1)$$

where $f_i(x)$ denotes the i -th component of $F(x)$, e_j the j -th unit vector, h a small perturbation of x . Equivalently, if $A_{.j}$ denotes the j -th column of A , we have

$$A_{.j} = \frac{F(x + he_j) - F(x)}{h}. \quad (3.4.2)$$

Theorem 3.4.1 *Let $F : R^n \rightarrow R^m$ satisfy the conditions of Theorem 1.2.22. Let the norm $\|\cdot\|$ satisfy $\|e_j\| = 1, j = 1, \dots, n$. Then*

$$\|A_{.j} - J(x)_{.j}\| \leq \frac{\gamma}{2}|h|. \quad (3.4.3)$$

If the norm used is l_1 norm, then

$$\|A - J(x)\|_1 \leq \frac{\gamma}{2}|h|. \quad (3.4.4)$$

Proof. By setting $d = he_j$ in (1.2.109), we obtain

$$\|F(x + he_j) - F(x) - J(x)he_j\| \leq \frac{\gamma}{2}\|he_j\|^2 = \frac{\gamma}{2}|h|^2.$$

Dividing by h gives (3.4.3). Noting from (1.2.7) that the l_1 norm of a matrix is the maximum of the l_1 norm of a vector, we immediately get (3.4.4). $\square$

Now, let $f : R^n \rightarrow R$. An approximation to the gradient $\nabla f(x)$ can be obtained by the forward-difference approximation, defined as

$$\frac{\partial f}{\partial x_i}(x) \approx \frac{f(x + he_i) - f(x)}{h}. \quad (3.4.5)$$

This process requires evaluation of f at $n + 1$ points: x and $x + he_i, i = 1, \dots, n$. Obviously, it follows from (1.2.109) that

$$\frac{\partial f}{\partial x_i}(x) = \frac{f(x + he_i) - f(x)}{h} + \delta_h, \quad (3.4.6)$$

where

$$|\delta_h| \leq \frac{\gamma}{2}h. \quad (3.4.7)$$

It means there is $O(h)$ error in the forward-difference formula.

A more accurate approximation to the derivative can be obtained by using the central-difference formula, defined as

$$\frac{\partial f}{\partial x_i}(x) \approx \frac{f(x + he_i) - f(x - he_i)}{2h}.$$

The two theorems below give respectively approximations to the gradient and the Hessian of f , and describe the error bounds of these approximations.

Theorem 3.4.2 *Let $f : D \subset R^n \rightarrow R$ satisfy the conditions of Theorem 1.2.23. Let the norm used satisfy $\|e_i\| = 1, i = 1, \dots, n$. Assume that $x + he_i, x - he_i \in D, i = 1, \dots, n$. Also let the vector $a \in R^n$ with components a_i , be defined as*

$$a_i = \frac{f(x + he_i) - f(x - he_i)}{2h}. \quad (3.4.8)$$

Then

$$|a_i - |\nabla f(x)|_i| \leq \frac{\gamma}{6}h^2. \quad (3.4.9)$$

If the norm used is the l_∞ norm, then

$$\|a - \nabla f(x)\|_\infty \leq \frac{\gamma}{6}h^2. \quad (3.4.10)$$

Proof. Define α and β respectively as

$$\alpha = f(x + he_i) - f(x) - h[\nabla f(x)]_i - \frac{1}{2}h^2[\nabla^2 f(x)]_{ii} \quad (3.4.11)$$

and

$$\beta = f(x - he_i) - f(x) + h[\nabla f(x)]_i - \frac{1}{2}h^2[\nabla^2 f(x)]_{ii}. \quad (3.4.12)$$

By using (1.2.110) and setting $d = \pm he_i$, we have

$$|\alpha| \leq \frac{\gamma}{6}h^3, \quad |\beta| \leq \frac{\gamma}{6}h^3.$$

Then using the triangle inequality gives

$$|\alpha - \beta| \leq \frac{\gamma}{3}h^3.$$

Also, from (3.4.11)-(3.4.12) and (3.4.8), we get

$$\alpha - \beta = 2h(a_i - [\nabla f(x)]_i),$$

which gives (3.4.9). Finally, by using the definition of l_∞ norm, we get (3.4.10) immediately from (3.4.9). $\square$

Theorem 3.4.3 *Let f satisfy the conditions of Theorem 3.4.2. Assume that $x, x + he_i, x + he_j, x + he_i + he_j \in D, 1 \leq i, j \leq n$. Also let $A \in R^{n \times n}$ with components a_{ij} defined as*

$$a_{ij} = \frac{f(x + he_i + he_j) - f(x + he_i) - f(x + he_j) + f(x)}{h^2}. \quad (3.4.13)$$

Then

$$|a_{ij} - [\nabla^2 f(x)]_{ij}| \leq \frac{5}{3}\gamma h. \quad (3.4.14)$$

If the matrix norm is l_1, l_∞ , or Frobenius norm, then

$$\|A - \nabla^2 f(x)\| \leq \frac{5}{3}\gamma hn. \quad (3.4.15)$$

Proof. The proof is similar to the proof in Theorem 3.4.2. Set

$$\begin{aligned} \alpha &= f(x + he_i + he_j) - f(x) - (he_i + he_j)^T \nabla f(x) \\ &\quad - \frac{1}{2}(he_i + he_j)^T \nabla^2 f(x)(he_i + he_j), \\ \beta &= f(x + he_i) - f(x) - (he_i)^T \nabla f(x) - \frac{1}{2}(he_i)^T \nabla^2 f(x)(he_i), \\ \eta &= f(x + he_j) - f(x) - (he_j)^T \nabla f(x) - \frac{1}{2}(he_j)^T \nabla^2 f(x)(he_j), \end{aligned}$$

respectively. Then

$$\alpha - \beta - \eta = h^2(a_{ij} - [\nabla^2 f(x)]_{ij}). \quad (3.4.16)$$

Also, we have

$$\begin{aligned} |\alpha - \beta - \eta| &\leq |\alpha| + |\beta| + |\eta| \\ &\leq \frac{\gamma}{6} \|he_i + he_j\|^3 + \frac{\gamma}{6} \|he_i\|^3 + \frac{\gamma}{6} \|he_j\|^3 \\ &\leq \frac{5}{3} \gamma h^3. \end{aligned}$$

This inequality together with (3.4.16) gives the result (3.4.14). The inequality (3.4.15) is a consequence of (3.4.14) and definitions of norms. $\square$

Now we are in a position to discuss the finite-difference Newton's method for nonlinear equations

$$F(x) = 0, \quad (3.4.17)$$

where $F : R^n \rightarrow R^n$ is continuously differentiable.

The Newton's method for (3.4.17) is as follows:

Solve $J(x_k)d = -F(x_k)$ for d_k ;

Set $x_{k+1} = x_k + \alpha_k d_k$;

where $J(x_k)$ is the Jacobian matrix of F at x_k . When $J(x)$ is not available, we can use finite-difference derivative approximation and get the following finite-difference Newton's method for (3.4.17):

$$(A_k)_{.j} = \frac{F(x_k + h_k e_j) - F(x_k)}{h_k}, \quad j = 1, \dots, n, \quad (3.4.18)$$

$$x_{k+1} := x_k - A_k^{-1} F(x_k), \quad k = 0, 1, \dots. \quad (3.4.19)$$

Theorem 3.4.4 *Let $F : R^n \rightarrow R^n$ be continuously differentiable on an open convex set $D \subset R^n$. Assume there exist $x^* \in R^n$ and $r, \beta > 0$, so that $N(x^*, r) \subset D$, $F(x^*) = 0$, $J(x^*)^{-1}$ exists and satisfies $\|J(x^*)^{-1}\| \leq \beta$, where J is Lipschitz continuous in the neighborhood $N(x^*, r) = \{x \in R^n \mid \|x - x^*\| < r\}$. Then there exist $\epsilon, h > 0$, such that if $x_0 \in N(x^*, \epsilon)$ and $\{h_k\}$ is a real sequence with $0 < |h_k| \leq h$, then the sequence $\{x_k\}$ generated from (3.4.18)-(3.4.19) is well-defined and converges to x^* linearly. If*

$$\lim_{k \rightarrow \infty} h_k = 0,$$

the convergence is superlinear. Furthermore, if there exists a constant c_1 , such that

$$|h_k| \leq c_1 \|x_k - x^*\|, \quad (3.4.20)$$

or equivalently, there exists a constant c_2 , such that

$$|h_k| \leq c_2 \|F(x_k)\|, \quad (3.4.21)$$

then the convergence rate is quadratic.

Proof. Choose ϵ and h such that, for $x_k \in N(x^*, \epsilon)$, A_k is nonsingular and $|h_k| < h$. Let $\epsilon \leq r$ and

$$\epsilon + h \leq \frac{1}{2\beta\gamma}. \quad (3.4.22)$$

Now we prove, by induction, that

$$\|x_{k+1} - x^*\| \leq \frac{1}{2} \|x_k - x^*\|, \quad (3.4.23)$$

so

$$x_{k+1} \in N(x^*, \epsilon). \quad (3.4.24)$$

For $k = 0$, we first prove A_0 is nonsingular. By assumptions and Theorem 3.4.1, we have $\|A(x) - J(x)\| \leq \frac{\gamma h}{2}$, and then

$$\begin{aligned} & \|J(x^*)^{-1}[A_0 - J(x^*)]\| \\ & \leq \|J(x^*)^{-1}\| \| [A_0 - J(x_0)] + [J(x_0) - J(x^*)] \| \\ & \leq \beta \left(\frac{\gamma h}{2} + \gamma \epsilon \right) \leq \frac{1}{2}. \end{aligned} \quad (3.4.25)$$

From Von-Neumann Theorem 1.2.5 we know that A_0 is nonsingular and that

$$\|A_0^{-1}\| \leq 2\beta. \quad (3.4.26)$$

Hence x_1 is well-defined and

$$\begin{aligned} x_1 - x^* &= -A_0^{-1}F(x_0) + x_0 - x^* \\ &= A_0^{-1}\{[F(x^*) - F(x_0) - J(x_0)(x^* - x_0)] \\ &\quad + [(J(x_0) - A_0)(x^* - x_0)]\}. \end{aligned} \quad (3.4.27)$$

Then from (3.4.26), (1.2.109) and (3.4.22), we get

$$\begin{aligned} \|x_1 - x^*\| &\leq \|A_0^{-1}\| \{ \|F(x^*) - F(x_0) - J(x_0)(x^* - x_0)\| \\ &\quad + \|A_0 - J(x_0)\| \|x^* - x_0\| \} \end{aligned} \quad (3.4.28)$$

$$\leq 2\beta \left\{ \frac{\gamma}{2} \|x^* - x_0\|^2 + \frac{\gamma}{2} h \|x_0 - x^*\| \right\} \quad (3.4.29)$$

$$\begin{aligned} &\leq \beta\gamma(\epsilon + h) \|x^* - x_0\| \\ &\leq \frac{1}{2} \|x_0 - x^*\|. \end{aligned} \quad (3.4.30)$$

Assume that the conclusion holds for $k = j$, in the same way as $k = 0$, we can prove that the conclusion is also true for $k = j + 1$. Therefore, (3.4.23)-(3.4.24) hold. They also show the linear convergence of the iterative sequence.

The key for superlinear and quadratic convergence requires an improved bound on $\|A_0 - J(x_0)\|$. When $\lim_{k \rightarrow \infty} h_k = 0$, the second term in the bracket of (3.4.29) approaches zero, and hence

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \rightarrow 0, \text{ when } k \rightarrow \infty,$$

which implies that the method converges superlinearly. Similarly, when (3.4.20) is satisfied, it follows from (3.4.29) that the method converges to x^* quadratically. Finally, the equivalence of (3.4.20) and (3.4.21) is just a consequence of Theorem 1.2.25. $\square$

For unconstrained optimization problem

$$\min_{x \in R^n} f(x), \quad (3.4.31)$$

when the gradient $\nabla f(x)$ is available, we can obtain the Hessian approximation by using the forward-difference or central-difference of the gradient. In this case, the iteration scheme for the k -th step is as follows:

$$(A)_{.j} = \frac{\nabla f(x_k + h_j e_j) - \nabla f(x_k)}{h_j}, j = 1, \dots, n, \quad (3.4.32)$$

$$A_k = \frac{A + A^T}{2}, \quad (3.4.33)$$

$$x_{k+1} = x_k - A_k^{-1} \nabla f(x_k), \quad (3.4.34)$$

where

$$h_j = \sqrt{\eta} \max\{|x_j|, \tilde{x}_j\} \text{sign}(x_j), \quad (3.4.35)$$

$\tilde{x}_j$ is a typical estimation given by users, and η is a small number more than the machine accuracy.

If the standard assumptions of Theorem 3.4.4 hold, and if h_j satisfies

$$h_j = O(\|x_k - x^*\|),$$

this finite-difference Newton's method (3.4.34) maintains the quadratic convergence rate.

Sometimes, some algorithms require us to supply the Hessian matrix-vector product $\nabla^2 f(x)d$, where d is a given vector. Instead of (3.4.32), we can use

$$\nabla^2 f(x_k)d \approx \frac{\nabla f(x_k + hd) - \nabla f(x_k)}{h}, \quad (3.4.36)$$

which also has $O(h)$ approximation error. For obtaining this approximation, the cost is only evaluation of a single gradient at $x_k + hd$. However, the cost of (3.4.32) is evaluation of the gradient at $n + 1$ points x_k and $x_k + h_j e_j, j = 1, \dots, n$.

In the case that the gradient $\nabla f(x)$ is not available, we can only use the function values to approximate the Hessian. The expression (3.4.13) gives the Hessian approximation as follows:

$$(A_k)_{ij} = \frac{[f(x_k + h_i e_i + h_j e_j) - f(x_k + h_i e_i)] - [f(x_k + h_j e_j) - f(x_k)]}{h_i h_j},$$

where

$$h_j = \sqrt[3]{\eta} \max\{|x_j|, \tilde{x}_j\} \text{sign}(x_j)$$

or

$$h_j = (\tilde{\epsilon})^{1/3} x_j,$$

where $\tilde{\epsilon}$ is a machine accuracy. Using the forward-difference and central-difference, the gradient approximations are respectively

$$(\hat{g}_k)_j = \frac{f(x_k + h_j e_j) - f(x_k)}{h_j}, \quad j = 1, \dots, n \quad (3.4.37)$$

and

$$(\hat{g}_k)_j = \frac{f(x_k + h_j e_j) - f(x_k - h_j e_j)}{2h_j}, \quad j = 1, \dots, n. \quad (3.4.38)$$

Their approximation errors are $O(h_j)$ and $O(h_j^2)$ respectively. In this case, the finite-difference Newton's iteration is

$$x_{k+1} = x_k - A_k^{-1} \hat{g}_k, \quad (3.4.39)$$

where A_k and $\hat{g}_k$ are finite-difference approximations of $\nabla^2 f(x_k)$ and $\nabla f(x_k)$ respectively. Under the standard assumptions of Theorem 3.4.4, we have similarly

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \|A_k^{-1}\| \left(\frac{v}{2} \|x_k - x^*\|^2 + \|A_k - \nabla^2 f(x_k)\| \|x_k - x^*\| \right. \\ &\quad \left. + \|\hat{g}_k - \nabla f(x_k)\| \right). \end{aligned} \quad (3.4.40)$$

Note that there is an additional term $\|\hat{g}_k - \nabla f(x_k)\|$ than (3.4.28). If we want to get the quadratic convergence rate, it is obvious to require $\|\hat{g}_k - \nabla f(x_k)\| = O(\|x_k - x^*\|^2)$ which implies $h_j = O(\|x_k - x^*\|^2)$. Therefore, it tells us that, when using the central-difference, the iteration (3.4.39) possesses quadratic rate. If we use the forward-difference, the iteration has quadratic rate only when $h_j = O(\|x_k - x^*\|^2)$.

In general, the forward-difference scheme is practical. Although the error of the central-difference scheme is $O(h_j^2)$, as compared to the $O(h_j)$ error in forward-difference, the cost is about twice as much as that of the forward-difference. Hence, we use the central-difference scheme only for those problems which need higher accuracy. Stewart [323] gave a switch rule from forward difference to central difference. Finally, it should be mentioned that, if the gradient is available, it is better to make the best use of it.

3.5 Negative Curvature Direction Method

Another strategy for modifying Newton's method, the negative curvature direction method, is presented, because the modified Newton's methods described above are not adequate for the case in which the Hessian $\nabla^2 f(x_k)$ is indefinite and x_k is close to a saddle point.

Now, we first put forward the definition below.

Definition 3.5.1 *Let $f : R^n \rightarrow R$ be twice continuously differentiable on an open set $D \subset R^n$.*

- (i) *If $\nabla^2 f(x)$ has at least a negative eigenvalue, then x is said to be an indefinite point.*

(ii) If x is an indefinite point and d satisfies $d^T \nabla^2 f(x) d < 0$, then d is said to be a negative curvature direction of $f(x)$ at x .

(iii) If

$$s^T \nabla f(x) \leq 0, \quad d^T \nabla f(x) \leq 0, \quad d^T \nabla^2 f(x) d < 0,$$

then the vector pair (s, d) is said to be a descent pair at the indefinite point x . If x is not an indefinite point and satisfies

$$s^T \nabla f(x) < 0, \quad d^T \nabla f(x) \leq 0, \quad d^T \nabla^2 f(x) d = 0,$$

then the vector pair (s, d) is said to be a descent pair at x .

As an example of a descent pair, we can choose

$$\begin{aligned} s &= -\nabla f(x), \\ d &= \begin{cases} 0, & \text{if } \nabla^2 f(x) \geq 0, \\ -\text{sign}(u^T \nabla f(x))u, & \text{otherwise,} \end{cases} \end{aligned}$$

where u is a unit eigenvector corresponding to a negative eigenvalue of $\nabla^2 f(x)$.

Obviously, there no longer exists the descent pair if and only if $\nabla f(x) = 0$ and $\nabla^2 f(x)$ is positive semi-definite.

From the definition above, at the stationary point, the negative curvature direction must be a descent direction. At a general point, if the negative curvature direction satisfies $d^T \nabla f(x) = 0$, then both d and $-d$ are descent directions. If $d^T \nabla f(x) \leq 0$, d is a descent direction, and if $d^T \nabla f(x) \geq 0$, $-d$ is a descent direction.

In this section, we first give the Gill-Murray stable Newton's method which uses negative curvature direction. Then we discuss two negative curvature direction methods: Fiacco-McCormick method and Fletcher-Freeman method. Finally, we consider the second order Armijo step rules and the second order Wolfe-Powell step rules.

3.5.1 Gill-Murray Stable Newton's Method

The basic idea of Gill-Murray stable Newton's method is: when the Hessian G_k is indefinite, one uses the modified Cholesky factorization to force the matrix G_k to be positive definite; when x_k approaches to a stationary point, use the negative curvature direction to decrease the objective function.

Let the modified Cholesky factorization be

$$\bar{G}_k = G_k + E_k = L_k D_k L_k^T,$$

where

$$D_k = \text{diag}(d_{11}, \dots, d_{nn}), \quad E_k = \text{diag}(e_{11}, \dots, e_{nn}).$$

When $\|g_k\| \leq \epsilon$ and $\nabla^2 f(x_k)$ is not positive semi-definite, we use the following negative curvature direction algorithm.

Algorithm 3.5.2

Step 1. Set $\psi_j = d_{jj} - e_{jj}$, $j = 1, \dots, n$.

Step 2. Find the subscript t , such that $\psi_t = \min\{\psi_j \mid j = 1, \dots, n\}$.

Step 3. If $\psi_t \geq 0$, stop; otherwise, solve

$$L_k^T d = e_t \tag{3.5.1}$$

for d_k , where e_t is a unit vector with the t -th component of e_t being 1. $\square$

Theorem 3.5.3 *Let G_k be the Hessian of $f(x)$ at x_k and*

$$\bar{G}_k = G_k + E_k = L_k D_k L_k^T.$$

If the direction d_k is obtained by Algorithm 3.5.2, then d_k is a negative curvature direction at x_k , and at least one in d_k and $-d_k$ is descent direction at x_k .

Proof. Since L_k is a unit lower triangular matrix, the solution d_k of (3.5.1) has the form

$$d_k = (\rho_1, \dots, \rho_{t-1}, 1, 0, \dots, 0)^T.$$

Then

$$\begin{aligned} d_k^T G_k d_k &= d_k^T \bar{G}_k d_k - d_k^T E_k d_k \\ &= d_k^T L_k D_k L_k^T d_k - d_k^T E_k d_k \\ &= e_t^T D_k e_t - \left(\sum_{r=1}^{t-1} \rho_r^2 e_{rr} + e_{tt} \right) \\ &= d_{tt} - e_{tt} - \sum_{r=1}^{t-1} \rho_r^2 e_{rr} \\ &= \psi_t - \sum_{r=1}^{t-1} \rho_r^2 e_{rr}. \end{aligned}$$

By the modified Cholesky factorization Algorithm 3.3.2, we have

$$\begin{aligned} e_{jj} &= \bar{g}_{jj} - g_{jj} = d_{jj} + \sum_{r=1}^{j-1} l_{jr}^2 d_r - g_{jj} \\ &= d_{jj} - c_{jj} \geq 0, \end{aligned}$$

which indicates that $\sum_{r=1}^{t-1} \rho_r^2 e_{rr} \geq 0$. Also, since $\psi_t < 0$, we obtain $d_k^T G_k d_k < 0$, which means d_k is a negative curvature direction, and $-d_k$ too. If $g_k^T d_k \leq 0$, then d_k is a descent direction; otherwise, $-d_k$ is a descent direction. $\square$

The algorithm below is the Gill-Murray numerically stable Newton's method.

Algorithm 3.5.4

Step 1. Given a starting point x_0 , $\epsilon > 0$. Set $k := 1$.

Step 2. Compute g_k and G_k .

Step 3. Compute modified Cholesky factorization by using Algorithm 3.3.2

$$G_k + E_k = L_k D_k L_k^T.$$

Step 4. If $\|g_k\| > \epsilon$, solve $L_k D_k L_k^T d_k = -g_k$ for d_k , and go to Step 6; otherwise, go to Step 5.

Step 5 Perform Algorithm 3.5.2. If it cannot produce d_k (i.e., $\psi_t \geq 0$), stop; otherwise, find d_k and set

$$d_k = \begin{cases} -d_k, & \text{if } g_k^T d_k > 0, \\ d_k, & \text{otherwise.} \end{cases}$$

Step 6. Compute line search factor α_k , and set $x_{k+1} = x_k + \alpha_k d_k$.

Step 7. If $f(x_{k+1}) \geq f(x_k)$, stop; otherwise, set $k = k + 1$, and go to Step 2. $\square$

About the convergence of the algorithm above, we have the following theorem.

Theorem 3.5.5 *Let $f : R^n \rightarrow R$ be twice continuously differentiable on an open set D . Assume there exists $\bar{x} \in D \subset R^n$ such that the level set*

$$L(\bar{x}) = \{x \mid f(x) \leq f(\bar{x})\}$$

is a bounded closed convex set. Assume that we pick $\epsilon = 0$ in Algorithm 3.5.4, and the starting point $x_0 \in L(\bar{x})$. Then the sequence $\{x_k\}$ generated from Algorithm 3.5.4 satisfies

- (i) *when $\{x_k\}$ is a finite sequence, its last element must be the stationary point of $f(x)$;*
- (ii) *when $\{x_k\}$ is an infinite sequence, it must have accumulation points, and all accumulation points are the stationary points of $f(x)$.*

The proof is omitted. We refer the interested reader to the original paper Gill and Murray [147].

3.5.2 Fiacco-McCormick Method

The idea of the negative curvature direction method was first presented by Fiacco and McCormick [122] who dealt with the case that the Hessian G_k has negative eigenvalues and employed the exact line search. The idea is simply to go forward along a negative curvature direction and decrease the objective function.

When

$$d_k^T g_k \leq 0 \text{ and } d_k^T G_k d_k < 0, \quad (3.5.2)$$

$$f(x_k + d_k) \approx f(x_k) + d_k^T g_k + \frac{1}{2} d_k^T G_k d_k$$

will be descending. Since G_k is indefinite, the Fiacco-McCormick method uses the decomposition

$$G_k = LDL^T, \quad (3.5.3)$$

where L is a unit lower triangular, and D is a diagonal matrix. If G_k is positive definite, the d_k generated from this decomposition (3.5.3) is a descent direction. However, if there exists a negative d_{ii} , then solve

$$L^T t = a, \quad (3.5.4)$$

where the components a_i of the vector a is defined as

$$a_i = \begin{cases} 1, & d_{ii} \leq 0, \\ 0, & d_{ii} > 0. \end{cases} \quad (3.5.5)$$

It is easy to show that

$$d_k = \begin{cases} t, & g_k^T t \leq 0, \\ -t, & g_k^T t > 0, \end{cases} \quad (3.5.6)$$

is a negative curvature direction satisfying (3.5.2).

Unfortunately, the decomposition (3.5.3) may be potentially unstable, amplify the rounding errors, and even do not exist. Hence, Fletcher and Freeman [135] employ a stable symmetric indefinite factorization.

3.5.3 Fletcher-Freeman Method

Fletcher and Freeman [135], instead, employ a stable symmetric indefinite factorization due to Bunch and Parlett [33]. For any symmetric matrix G_k , there exists a permutation matrix, such that

$$P^T G_k P = LDL^T, \quad (3.5.7)$$

where L is unit lower triangular, D is a block diagonal matrix with blocks of dimension 1 or 2. The aim to use the permutation matrix is to maintain the symmetricity and numerical stability. Contrasting with the factorization (3.5.3), the factorization (3.5.7) always exists and can be computed by a numerically stable process. Now, for 1×1 pivot case, let A be an $n \times n$ matrix

$$A = A^{(0)} = \begin{bmatrix} a_{11} & \vec{a}_{21}^T \\ \vec{a}_{21} & A_{22} \end{bmatrix}, \quad (3.5.8)$$

where $\vec{a}_{21}$ is $(n-1) \times 1$ vector, A_{22} is an $(n-1) \times (n-1)$ matrix. Eliminating one row and one column yields a reduced matrix $A^{(1)}$:

$$A^{(1)} = A^{(0)} - d_{11} l_1 l_1^T = \begin{bmatrix} 0 & 0^T \\ 0 & A_{22} - \vec{a}_{21} \vec{a}_{21}^T / d_{11} \end{bmatrix}, \quad (3.5.9)$$

where

$$d_{11} = a_{11}, \quad l_1 = \frac{1}{d_{11}} \begin{bmatrix} a_{11} \\ \vec{a}_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ \vec{a}_{21}/d_{11} \end{bmatrix}. \quad (3.5.10)$$

For 2×2 pivot case, let

$$A^{(0)} = \begin{bmatrix} A_{11} & A_{21}^T \\ A_{21} & A_{22} \end{bmatrix}, \quad (3.5.11)$$

where A_{11} is a 2×2 block matrix, A_{21} is an $(n-2) \times 2$ matrix, and A_{22} is an $(n-2) \times (n-2)$ matrix. Eliminating two rows and two columns yields a reduced matrix $A^{(2)}$:

$$\begin{aligned} A^{(2)} &= A^{(0)} - L_1 D_1 L_1^T = A^{(0)} - \begin{bmatrix} I \\ L_{21} \end{bmatrix} D_1 [I \ L_{21}^T] \\ &= \begin{bmatrix} 0 & 0 \\ 0 & A_{22} - A_{21} D_1^{-1} A_{21}^T \end{bmatrix}, \end{aligned} \quad (3.5.12)$$

where

$$D_1 = A_{11}, \quad L_1 = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} D_1^{-1} = \begin{bmatrix} I \\ A_{21} A_{11}^{-1} \end{bmatrix} \triangleq \begin{bmatrix} I \\ L_{21} \end{bmatrix}. \quad (3.5.13)$$

Next step, we will apply the same process to the remaining matrix $A_{22} - \tilde{a}_{21} \tilde{a}_{21}^T / d_{11}$ or $A_{22} - A_{21} D_1^{-1} A_{21}^T$ with dimension $(n-1) \times (n-1)$ or $(n-2) \times (n-2)$ respectively. Finally, this recursive procedure gives (3.5.7).

In all the iterations, the algorithm has to identify the pivot block between two pivoting forms. A natural problem is how to identify 1×1 submatrix a_{11} or 2×2 block submatrix A_{11} as a pivot block. Now we describe a criteria as follows. First, compute the largest-magnitude diagonal and the largest-magnitude off-diagonal elements, denoting their respective magnitude by ξ_{dia} and ξ_{off} . If the growth ratio ξ_{dia}/ξ_{off} is acceptable, we choose the diagonal element with largest-magnitude as a pivot and perform row-column exchange such that a_{11} is just the element. Otherwise, we choose the off-diagonal element, say a_{ij} , whose magnitude is ξ_{off} , and choose the corresponding 2×2 block

$$\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ij} & a_{jj} \end{bmatrix}$$

as a pivot block. Then we perform row-column exchange such that A_{11} is this 2×2 block.

This decomposition needs $n^3/6 + O(n^2)$ multiplications. Maybe the expensive computation is a disadvantage of this method. A more economical improvement is presented by Bunch and Kaufman [32]. The interested reader

may consult that paper. The forms of L and D produced by the decomposition are a block lower triangular matrix and a block diagonal matrix, for example,

$$D = \begin{bmatrix} * & & & & \\ & * & * & & \\ & * & * & & \\ & & & * & * \\ & & & * & * \\ & & & & * \end{bmatrix}, \quad L = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ 0 & 1 & & & \\ * & * & 1 & & \\ * & * & 0 & 1 & \\ * & * & * & * & 1 \end{bmatrix}.$$

The decomposition above is said to be Bunch-Parlett factorization, in brief, B-P factorization which can generate negative curvature direction.

Let G_k have symmetric indefinite factorization

$$G_k = LDL^T. \quad (3.5.14)$$

We solve the triangular system of equations

$$L^T t = a, \quad (3.5.15)$$

where, in the case of 1×1 pivot, the components of a are

$$a_i = \begin{cases} 1, & d_{ii} \leq 0, \\ 0, & d_{ii} > 0; \end{cases} \quad (3.5.16)$$

in the case of 2×2 pivot, $\begin{pmatrix} a_i \\ a_{i+1} \end{pmatrix}$ is the unit eigenvector corresponding to the negative eigenvalue of $\begin{bmatrix} d_{ii} & d_{i,i+1} \\ d_{i+1,i} & d_{i+1,i+1} \end{bmatrix}$. Set

$$d_k = \begin{cases} t, & \text{when } g_k^T t \leq 0, \\ -t, & \text{when } g_k^T t > 0, \end{cases} \quad (3.5.17)$$

then d_k is the negative curvature direction satisfying (3.5.2). In fact, we have

$$d_k^T G_k d_k = d_k^T LDL^T d_k = a^T D a = \sum_{i: \lambda_i < 0} \lambda_i < 0, \quad (3.5.18)$$

and

$$d_k^T g_k \leq 0. \quad (3.5.19)$$

In addition, when D has negative eigenvalues, the direction d_k can also be computed by

$$d_k = -L^{-T} \tilde{D}^+ L^{-1} g_k, \quad (3.5.20)$$

where $\tilde{D}$ is the positive part of D , i.e.,

$$\tilde{D}_i = \begin{cases} d_{ii}, & \text{when } d_{ii} > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and $\tilde{D}^+$ is the generalized inverse of $\tilde{D}$.

When D contains at least one zero eigenvalue, the direction d_k can be computed by

$$G_k d_k = L D L^T d_k = 0, \quad g_k^T d_k < 0. \quad (3.5.21)$$

When all the eigenvalues of D are positive, all blocks of D are 1×1 elements. In this case, B-P decomposition is reduced to usual Cholesky factorization, and the direction produced is usual Newton's direction

$$d_k = -L^{-T} D^{-1} L^{-1} g_k.$$

It is not difficult to see that the negative curvature descent direction determined by (3.5.17) is limited in some subspace; the direction from (3.5.20) is a Newton's direction limited in the subspace of positive curvature direction. Although the idea of using negative curvature directions is in some ways attractive, Fletcher and Freeman [135] find that it is not satisfactory to use such directions on successive iterations and that if we alternate positive curvature and negative curvature search, i.e., alternate (3.5.17) and (3.5.20), we can get better results. Similarly, if one continuously meets zero eigenvalue, alternating (3.5.20) and (3.5.21) will give better results.

3.5.4 Second-Order Step Rules

Second-Order Armijo Step Rule – McCormick Method

In §2.5 we have discussed Armijo line search rule. Consider

$$\min f(x), \quad x \in \mathcal{D} \subset R^n, \quad (3.5.22)$$

where $f : R^n \rightarrow R$ is a continuously differentiable function in the open set $\mathcal{D}$.

Given $\beta \in (0, 1)$ and $\rho \in (0, 1)$, m_k is the least nonnegative integer m such that

$$f(x_k + \beta^m \tau d_k) \leq f(x_k) + \rho \beta^m \tau g_k^T d_k, \quad (3.5.23)$$

where $\tau > 0$, or require α to satisfy

$$f(x_k + \alpha d_k) \leq f(x_k) + \rho \alpha g_k^T d_k. \quad (3.5.24)$$

For the steepest descent method

$$x_{k+1} = x_k - 2^{-i} g_k, \quad (3.5.25)$$

the Armijo rule is

$$f(x_{k+1}) \leq f(x_k) - \rho 2^{-i} \|g_k\|^2, \quad \rho \in (0, 1). \quad (3.5.26)$$

Instead of using only one descent direction and searching in a line determined by that direction, we search along a curve of the form

$$x(\alpha) = x_k + \phi_1(\alpha) s_k + \phi_2(\alpha) d_k, \quad (3.5.27)$$

where (s_k, d_k) is a descent pair at x_k defined in Definition 3.5.1, $\phi_1(\alpha)$ and $\phi_2(\alpha)$ are polynomials with $\phi_1(0) = \phi_2(0) = 0$.

If we set $\Phi(\alpha) = f(x(\alpha))$ and assume that $\rho \in (0, 1)$, there is an $\bar{\alpha} > 0$ such that

$$\Phi(\alpha) \leq \Phi(0) + \rho[\Phi'(0)\alpha + \frac{1}{2}\Phi''(0)\alpha^2] \quad (3.5.28)$$

for all $\alpha \in [0, \bar{\alpha}]$ provided that either $\Phi'(0) < 0$ or $\Phi'(0) = 0$ and $\Phi''(0) < 0$.

Normally, in (3.5.27) we choose $\phi_1(\alpha)$ and $\phi_2(\alpha)$ as lower-order polynomials. The simplest functions of this type are

$$\phi_1(\alpha) = \alpha^2, \quad \phi_2(\alpha) = \alpha,$$

which lead to the iteration

$$x(\alpha) = x_k + \alpha^2 s_k + \alpha d_k. \quad (3.5.29)$$

If we set $\alpha = \gamma^i$, $\gamma \in (0, 1)$, (3.5.29) becomes

$$x_k(i) = x_k + \gamma^{2i} s_k + \gamma^i d_k \in \mathcal{D}. \quad (3.5.30)$$

The second-order Armijo rule requires us to find $i(k)$ which is the smallest nonnegative integer i such that

$$f(x_k(i)) \leq f(x_k) + \rho\gamma^{2i}[g_k^T s_k + \frac{1}{2}d_k^T G_k d_k], \quad (3.5.31)$$

where $\rho \in (0, 1)$, and set $x_{k+1} = x_k(i(k))$. Typically, McCormick [203] chooses $\gamma^2 = \frac{1}{2}$ in (3.5.30).

There exists a finite $i(k)$ satisfying (3.5.31) provided that

$$s_k^T g_k < 0, \text{ whenever } g_k \neq 0 \quad (3.5.32)$$

and

$$d_k^T G_k d_k < 0, \text{ whenever } g_k = 0. \quad (3.5.33)$$

Only if x_k is a point satisfying the second-order optimal condition, there does not exist the descent pair satisfying (3.5.32)-(3.5.33), and the algorithm terminates. The following is the convergence theorem of the second-order Armijo rule.

Theorem 3.5.6 *Let $f : R^n \rightarrow R$ be twice continuously differentiable on the open set $\mathcal{D}$, and assume that for some $x_0 \in \mathcal{D}$, the level set*

$$L(x_0) = \{x \in \mathcal{D} : f(x) \leq f(x_0)\}$$

is compact. Suppose that $\{\|s_k\|\}$ and $\{\|d_k\|\}$ are bounded. If $\{x_k\}$ satisfies (3.5.30) and (3.5.31), then

$$\lim_{k \rightarrow \infty} g_k^T s_k = 0 \quad (3.5.34)$$

and

$$\lim_{k \rightarrow \infty} d_k^T G_k d_k = 0. \quad (3.5.35)$$

Proof. The sequence $\{f(x_k)\}$ is decreasing and bounded below due to the continuity of f and the compactness of $L(x_0)$. Thus $\{f(x_k) - f(x_{k+1})\}$ converges to zero. Let $i(k)$ be the smallest nonnegative integer such that (3.5.30)-(3.5.31) hold, then there are two cases to consider.

Case 1. Suppose the integer sequence $\{i(k)\}$ is bounded above by $\beta \geq 0$. Then

$$f(x_k) - f(x_{k+1}) \geq -\rho\gamma^{2\beta}[g_k^T s_k + \frac{1}{2}d_k^T G_k d_k]. \quad (3.5.36)$$

Since $-g_k^T s_k \geq 0$ and $-d_k^T G_k d_k \geq 0$, the conclusion follows.

Case 2. Suppose that the integer $\{i(k)\}$ is not bounded above. So, without loss of generality, we can assume that $\lim_{k \rightarrow +\infty} i(k) = +\infty$. By (3.5.30)-(3.5.31),

$$f(x(i(k) - 1)) - f(x_k) > \rho \gamma^{2[i(k)-1]} [g_k^T s_k + \frac{1}{2} d_k^T G_k d_k]. \quad (3.5.37)$$

For convenience, let

$$p_k = \gamma^{2[i(k)-1]} s_k + \gamma^{i(k)-1} d_k.$$

By using Taylor's theorem and noting that $\nabla^2 f(x)$ is continuous, we have

$$f(x(i(k) - 1)) - f(x_k) = p_k^T g_k + \frac{1}{2} p_k^T G_k p_k + o(\gamma^{2[i(k)-1]}). \quad (3.5.38)$$

Combining (3.5.37) and (3.5.38) gives

$$o(\gamma^{2[i(k)-1]}) > (1 - \rho) \gamma^{2[i(k)-1]} [-g_k^T s_k - \frac{1}{2} d_k^T G_k d_k]. \quad (3.5.39)$$

Dividing by $(1 - \rho) \gamma^{2[i(k)-1]}$ and taking limits yields

$$g_k^T s_k \rightarrow 0 \text{ and } d_k^T G_k d_k \rightarrow 0. \quad \square$$

Furthermore, we have the following result.

Theorem 3.5.7 *Assume that the conditions in Theorem 3.5.6 hold. In addition, suppose there exist positive constants c_1, c_2, c_3 , such that*

$$\|s_k\| \geq c_3 \|g_k\|, \quad (3.5.40)$$

$$d_k^T G_k d_k \leq c_2 \lambda_{G_k}, \quad (3.5.41)$$

$$-s_k^T g_k \geq c_1 \|s_k\| \|g_k\|, \quad (3.5.42)$$

where λ_{G_k} is the most negative eigenvalue of G_k . Then the accumulation point x^* of $\{x_k\}$ satisfies $\nabla f(x^*) = 0$, and $\nabla^2 f(x^*)$ is positive semi-definite with at least one zero eigenvalue.

Proof. From Theorem 3.5.6, we have

$$g_k^T s_k \rightarrow 0 \text{ and } d_k^T G_k d_k \rightarrow 0.$$

Using (3.5.42) and (3.5.40) gives

$$-g_k^T s_k \geq c_1 c_3 \|g_k\|^2.$$

Thus we get $\|g_k\| \rightarrow 0$. Also, it follows from (3.5.41) that $\bar{d}^T \nabla^2 f(x^*) \bar{d} = 0$ with $\bar{d}$ a limit eigenvector of a subsequence of $\{d_k\}$. Therefore, $\nabla^2 f(x^*)$ is positive semi-definite with at least one zero eigenvalue. $\square$

Second-Order Armijo Step Rule — Goldfarb Method

Goldfarb [154] thinks that the iteration

$$x_k(\alpha) = x_k + \alpha^2 s_k + \alpha d_k \quad (3.5.43)$$

is not ideal. The form (3.5.43) may be good in the neighborhood of a saddle point. However, far from a saddle point, it is not a good approach. Then Goldfarb [154] put forward a similar second-order Armijo rule based on the iteration of the form

$$x_k(\alpha) = x_k + \alpha s_k + \alpha^2 d_k, \quad (3.5.44)$$

and gives the following algorithm:

For given γ and ρ , where $0 < \gamma, \rho < 1$, and an initial point x_0 , determine x_{k+1} , for $k = 0, 1, \dots$, as follows:

Choose a descent pair (s_k, d_k) at x_k . If none exists, stop. Otherwise, let $i(k) + 1$ be the smallest nonnegative integer such that

$$f(x_k(\gamma^i)) - f(x_k) \leq \rho[\gamma^i s_k^T g_k + \frac{1}{2} \gamma^{4i} d_k^T G_k d_k] \quad (3.5.45)$$

and set

$$x_{k+1} = x_k(\gamma^{i(k)+1}). \quad (3.5.46)$$

In very much the same manner as Theorem 3.5.6 and Theorem 3.5.7, we have the convergence theorems. So we give them as follows without proof.

Theorem 3.5.8 *Let $f : R^n \rightarrow R$ have two continuous derivatives on the open set $\mathcal{D}$ and let the level set $S = \{x \mid f(x) \leq f(x_0)\}$ be a compact subset of $\mathcal{D}$ for a given $x_0 \in \mathcal{D}$. Suppose that an admissible sequence of descent pairs $\{(s_k, d_k)\}$ is used in the above algorithm, and that*

$$-s_k^T g_k \geq c_1 \|s_k\|^2, \quad (3.5.47)$$

$$s_k^T G_k s_k \leq c_2 \|s_k\|^2, \quad (3.5.48)$$

where $0 < c_1, c_2 < \infty$. Then $g_k \rightarrow 0$, $s_k \rightarrow 0$, $\lambda_k \rightarrow 0$, and $d_k \rightarrow 0$.

Theorem 3.5.9 *In addition to the assumptions of Theorem 3.5.8, assume that the set of stationary points of $f(x)$ in the level set L is finite. Then, if $\{x_k\}$ is the sequence obtained by the second-order Armijo steplength algorithm (3.5.45) and (3.5.46), we have*

$$\lim_{k \rightarrow \infty} x_k = x^*, \quad g(x^*) = 0, \quad G(x^*) \geq 0. \quad (3.5.49)$$

Moreover, if infinitely many $G_k \not\geq 0$, then $G(x^*)$ has at least one eigenvalue equal to zero.

Second-Order Wolfe-Powell Step Rule — Moré-Sorensen Rule

Consider the iteration of the form

$$x(\alpha) = x_k + \alpha^2 s_k + \alpha d_k, \quad (3.5.50)$$

where (s_k, d_k) is a descent pair at x_k . Replacing Wolfe-Powell step rule (2.5.3) and (2.5.7), we ask α to satisfy

$$f(x(\alpha)) \leq f(x) + \rho \alpha^2 [\nabla f(x)^T s + \frac{1}{2} d^T \nabla^2 f(x) d], \quad (3.5.51)$$

$$\nabla f(x(\alpha))^T x'(\alpha) \geq \sigma [\nabla f(x)^T d + 2\alpha \nabla f(x)^T s + \alpha d^T \nabla^2 f(x) d], \quad (3.5.52)$$

where $0 < \rho \leq \sigma < 1$. When $d = 0$, these conditions reduce to those of (2.5.3) and (2.5.7). The conditions (3.5.51) and (3.5.52) are said to be the second-order Wolfe-Powell step rule which is contributed by Moré and Sorensen [221].

If (s_k, d_k) is a descent pair at x_k and we set

$$\Phi_k(\alpha) = f(x_k + \alpha^2 s_k + \alpha d_k), \quad (3.5.53)$$

then (3.5.51) and (3.5.52) are equivalent to

$$\Phi_k(\alpha_k) \leq \Phi_k(0) + \frac{1}{2} \rho \Phi_k''(0) \alpha_k^2, \quad (3.5.54)$$

$$\Phi_k'(\alpha_k) \geq \sigma [\Phi_k'(0) + \Phi_k''(0) \alpha_k]. \quad (3.5.55)$$

The second order Wolfe-Powell step rule has a geometric interpretation as shown in Figure 3.5.1.

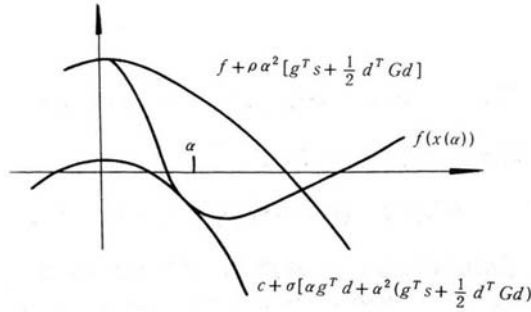


Figure 3.5.1 Second-order Wolfe-Powell step rule

Similar to the preceding discussion, we now give the following convergence results.

Theorem 3.5.10 *Let $f : R^n \rightarrow R$ have twice continuous derivatives on the open set $\mathcal{D}$, and assume that for some $x_0 \in \mathcal{D}$, the level set*

$$L(x_0) = \{x \in \mathcal{D} \mid f(x) \leq f(x_0)\}$$

is a compact subset of $\mathcal{D}$. If $\{x_k\}$ satisfies (3.5.50)-(3.5.52), then

$$\lim_{k \rightarrow \infty} g_k^T s_k = 0 \text{ and } \lim_{k \rightarrow \infty} d_k^T G_k d_k = 0. \quad (3.5.56)$$

Proof. From (3.5.53) we have $\Phi'_k(0) = g_k^T d_k$ and

$$\Phi''_k(0) = 2g_k^T s_k + d_k^T G_k d_k.$$

Since (s_k, d_k) is a descent pair, $\Phi'_k(0) \leq 0$ and $\Phi''_k(0) < 0$. Thus (3.5.51) implies that $\{x_k\} \subset L(x_0)$. By the continuity of f and compactness of $L(x_0)$ we have that $\{f_k - f_{k+1}\}$ converges to zero. Since

$$f_k - f_{k+1} \geq -\frac{1}{2} \rho \Phi''_k(0) \alpha_k^2 \geq 0,$$

it follows that

$$\lim_{k \rightarrow \infty} \alpha_k^2 g_k^T s_k = 0 \quad (3.5.57)$$

and

$$\lim_{k \rightarrow \infty} \alpha_k^2 d_k^T G_k d_k = 0. \quad (3.5.58)$$

From (3.5.55) we have

$$\Phi'_k(\alpha_k) - \Phi'_k(0) - \alpha_k \Phi''_k(0) \geq -(1 - \sigma)[\Phi'_k(0) + \Phi''_k(0)\alpha_k],$$

and hence

$$\Phi'_k(\alpha_k) - \Phi'_k(0) - \alpha_k \Phi''_k(0) \geq -(1 - \sigma)\Phi''_k(0)\alpha_k.$$

An application of the mean-value theorem yields that for some $\theta_k \in (0, \alpha_k)$,

$$\Phi''_k(\theta_k) - \Phi''_k(0) \geq -(1 - \sigma)\Phi''_k(0). \quad (3.5.59)$$

In the following, we prove (3.5.56) by contradiction. Suppose either the first equality or the second equality does not hold, then there is a subsequence $\{k_i\}$ and $\eta > 0$ such that

$$-\Phi''_{k_i}(0) \geq \eta > 0. \quad (3.5.60)$$

Hence (3.5.59) implies that $\{\alpha_{k_i}\}$ does not converge to zero. However, if $\{\alpha_{k_i}\}$ does not converge to zero and (3.5.60) holds, then (3.5.57) and (3.5.58) cannot be satisfied. This contradiction establishes the theorem. $\square$

Furthermore, we have

Theorem 3.5.11 *Let $f : R^n \rightarrow R$ have twice continuous derivatives on the open set $\mathcal{D}$, and assume that, for some $x_0 \in \mathcal{D}$, the level set $L(x_0) = \{x \in \mathcal{D} \mid f(x) \leq f(x_0)\}$ is compact. In addition, assume that f has a finite number of critical points in $L(x_0)$. Then, if $\{x_k\}$ is a sequence obtained by the second-order step rule (3.5.50)-(3.5.52), we have*

$$\lim_{k \rightarrow \infty} x_k = x^*, \quad g(x^*) = 0, \quad G(x^*) \geq 0. \quad (3.5.61)$$

Moreover, if infinitely many $G_k \not\geq 0$, then $G(x^*)$ has at least one eigenvalue equal to zero.

Proof. It is similar to the proof of Theorem 3.5.7. $\square$

Determine Descent Pair (s_k, d_k)

Finally, we mention a way to obtain the descent pair (s_k, d_k) which satisfies all of the requirements of Theorem 3.5.10 and 3.5.11. First, consider computing s_k . Assume that

$$G_k = L_k D_k L_k^T$$

is the Bunch-Parlett symmetric indefinite factorization where we omit the permutations, L_k is a unit lower triangular matrix, D_k a block diagonal matrix with 1×1 or 2×2 diagonal block. Let

$$D_k = U_k \Lambda_k U_k^T$$

be the spectral decomposition of D_k . Set

$$\bar{\lambda}_j^{(k)} = \max \left\{ |\lambda_j^{(k)}|, \epsilon n \max_{1 \leq i \leq n} |\lambda_i^{(k)}|, \epsilon \right\}, \quad j = 1, \dots, n,$$

$$\bar{\Lambda}_k = \text{diag}(\bar{\lambda}_1^{(k)}, \dots, \bar{\lambda}_n^{(k)}),$$

where ϵ is the relative machine precision. Set

$$\bar{D}_k = U_k \bar{\Lambda}_k U_k^T.$$

We obtain s_k as the solution of

$$L_k \bar{D}_k L_k^T s = -g_k.$$

Next, the negative curvature direction d_k is obtained as the solution of

$$L_k^T d_k = \pm |\min\{\lambda(D_k), 0\}|^{\frac{1}{2}} z_k,$$

where $\lambda(D_k)$ is the smallest eigenvalue of D_k and z_k the corresponding unit eigenvector of D_k . The other way to obtain a negative curvature direction d_k is to solve

$$L_k^T d_k = \pm \sum_{\lambda_j(D_k) \leq 0} z_j.$$

3.6 Inexact Newton's Method

As mentioned before, the pure Newton's method is expensive in each iteration, especially when the dimension n is large. Also, the quadratic model used to derive the Newton equation may not provide a good prediction of the behavior of the function, especially when the iterate x_k is remote from the solution x^* . In this section, we consider a class of inexact Newton's methods in which we only approximately solve the Newton equation. In the following, we discuss this class of methods for solving nonlinear equations $F(x) = 0$. It

is not difficult for readers to deal with unconstrained optimization problems by using this way.

Consider solving the nonlinear equations

$$F(x) = 0, \quad (3.6.1)$$

where $F : R^n \rightarrow R^n$ is assumed to have the following properties:

A1 There exists x^* such that $F(x^*) = 0$.

A2 F is continuously differentiable in the neighborhood of x^* .

A3 $F'(x^*)$ is nonsingular.

Recall that the basic Newton's step is obtained by solving

$$F'(x_k)s_k = -F(x_k) \quad (3.6.2)$$

and setting

$$x_{k+1} = x_k + s_k. \quad (3.6.3)$$

Now, we consider inexact Newton's method: solve

$$F'(x_k)s_k = -F(x_k) + r_k, \quad (3.6.4)$$

where

$$\|r_k\| \leq \eta_k \|F(x_k)\|. \quad (3.6.5)$$

Set

$$x_{k+1} = x_k + s_k. \quad (3.6.6)$$

Here, $r_k = F'(x_k)s_k + F(x_k)$ denotes the residual, and $\{\eta_k\}$ (with $0 < \eta_k < 1$) is a forcing sequence which controls the inexactness.

Next, we study the local convergence of inexact Newton's methods.

Lemma 3.6.1 *Let $F : D \subset R^n \rightarrow R^n$ be continuously differentiable in a neighborhood of $x^* \in D$, and let $F'(x^*)$ be nonsingular. Then there exist $\delta > 0, \xi > 0$, and $\epsilon > 0$, such that when $\|y - x^*\| < \delta$ and $y \in D$, $F'(y)$ is nonsingular and*

$$\|F'(y)^{-1}\| \leq \xi. \quad (3.6.7)$$

Also, $F'(y)^{-1}$ is continuous at x^ , that is*

$$\|F'(y)^{-1} - F'(x^*)^{-1}\| < \epsilon. \quad (3.6.8)$$

Proof. Set $\alpha = \|F'(x^*)^{-1}\|$. For a given $\beta < \alpha^{-1}$, choose δ such that when $\|y - x^*\| < \delta$ with $y \in D$,

$$\|F'(x^*) - F'(y)\| \leq \beta.$$

It follows from Von-Neumann Theorem 1.2.5 that $F'(y)$ is invertible, and (3.6.7) holds with $\xi = \alpha/(1 - \beta\alpha)$. Thus,

$$\begin{aligned} \|F'(x^*)^{-1} - F'(y)^{-1}\| &= \|F'(x^*)^{-1}(F'(y) - F'(x^*))F'(y)^{-1}\| \\ &\leq \alpha\xi\|F'(x^*) - F'(y)\| \\ &\leq \alpha\beta\xi \\ &\triangleq \epsilon, \end{aligned}$$

which says that the continuity of F' guarantees the continuity of $(F')^{-1}$. $\square$

In the following, we establish the linear convergence in Theorem 3.6.2 and superlinear convergence in Theorem 3.6.4.

Theorem 3.6.2 *Let $F : R^n \rightarrow R^n$ satisfy the properties (A1)–(A3). Assume that the sequence $\{\eta_k\}$ satisfies $0 \leq \eta_k \leq \eta < t < 1$. Then, for some $\epsilon > 0$, if the starting point x_0 is sufficiently near x^* , the sequence $\{x_k\}$ generated by inexact Newton's method (3.6.4)–(3.6.6) converges to x^* , and the convergence rate is linear, i.e.,*

$$\|x_{k+1} - x^*\|_* \leq t\|x_k - x^*\|_*, \quad (3.6.9)$$

where $\|y\|_* = \|F'(x^*)y\|$.

Proof. Since $F'(x^*)$ is nonsingular, for $y \in R^n$, we have

$$\frac{1}{\mu}\|y\| \leq \|y\|_* \leq \mu\|y\|, \quad (3.6.10)$$

where

$$\mu = \max\{\|F'(x^*)\|, \|F'(x^*)^{-1}\|\}. \quad (3.6.11)$$

Since $\eta < t$, there exists sufficiently small $\gamma > 0$, such that

$$(1 + \gamma\mu)[\eta(1 + \mu\gamma) + 2\mu\gamma] \leq t. \quad (3.6.12)$$

Now choose $\epsilon > 0$ sufficiently small, such that if $\|y - x^*\| \leq \mu^2\epsilon$, we have

$$\|F'(y) - F'(x^*)\| \leq \gamma, \quad (3.6.13)$$

$$\|F'(y)^{-1} - F'(x^*)^{-1}\| \leq \gamma, \quad (3.6.14)$$

$$\|F(y) - F(x^*) - F'(x^*)(y - x^*)\| \leq \gamma\|y - x^*\|. \quad (3.6.15)$$

Let $\|x_0 - x^*\| \leq \epsilon$. We now prove (3.6.9) by induction. By using (3.6.10)–(3.6.11) and assumption of the induction, we have

$$\begin{aligned} \|x_k - x^*\| &\leq \mu \|x_k - x^*\|_* \leq \mu t^k \|x_0 - x^*\|_* \\ &\leq \mu^2 \|x_0 - x^*\| \leq \mu^2 \epsilon. \end{aligned}$$

Then, when $y = x_k$, (3.6.13)–(3.6.15) hold. Since

$$\begin{aligned} &F'(x^*)(x_{k+1} - x^*) \\ = &F'(x^*)(x_k - x^* - F'(x_k)^{-1}F(x_k) + F'(x_k)^{-1}r_k) \\ = &F'(x^*)F'(x_k)^{-1}[F'(x_k)(x_k - x^*) - F(x_k) + r_k] \\ = &[I + F'(x^*)(F'(x_k)^{-1} - F'(x^*)^{-1})][r_k + (F'(x_k) - F'(x^*))(x_k - x^*) \\ &\quad - (F(x_k) - F(x^*) - F'(x^*)(x_k - x^*))], \end{aligned} \quad (3.6.16)$$

by taking norms and using (3.6.11), (3.6.14), (3.6.5), (3.6.13) and (3.6.15), we obtain

$$\begin{aligned} &\|x_{k+1} - x^*\|_* \\ \leq &[1 + \|F'(x^*)\| \|F'(x_k)^{-1} - F'(x^*)^{-1}\|][\|r_k\| + \\ &\|F'(x_k) - F'(x^*)\| \|x_k - x^*\| + \|F(x_k) - F(x^*) - F'(x^*)(x_k - x^*)\|] \\ \leq &(1 + \mu\gamma)[\eta_k \|F(x_k)\| + \gamma \|x_k - x^*\| + \gamma \|x_k - x^*\|]. \end{aligned} \quad (3.6.17)$$

Note that

$$F(x_k) = [F'(x^*)(x_k - x^*)] + [F(x_k) - F(x^*) - F'(x^*)(x_k - x^*)],$$

taking the norm gives

$$\|F(x_k)\| \leq \|x_k - x^*\|_* + \gamma \|x_k - x^*\|. \quad (3.6.18)$$

Substituting (3.6.18) into (3.6.17) and using (3.6.10) and (3.6.12) yield

$$\begin{aligned} \|x_{k+1} - x^*\|_* &\leq (1 + \mu\gamma)[\eta_k(\|x_k - x^*\|_* + \gamma \|x_k - x^*\|) + 2\gamma \|x_k - x^*\|] \\ &\leq (1 + \mu\gamma)[\eta(1 + \mu\gamma) + 2\mu\gamma]\|x_k - x^*\|_* \\ &\leq t\|x_k - x^*\|_*. \quad \square \end{aligned}$$

Below, we discuss the superlinear convergence rate of the inexact Newton's methods. We first give a lemma.

Lemma 3.6.3 *Let*

$$\alpha = \max\{\|F'(x^*)\| + \frac{1}{2\beta}, 2\beta\},$$

where $\beta = \|F'(x^*)^{-1}\|$. Then, for $\|y - x^*\|$ sufficiently small, the inequality

$$\frac{1}{\alpha}\|y - x^*\| \leq \|F(y)\| \leq \alpha\|y - x^*\| \quad (3.6.19)$$

holds.

Proof. From the continuous differentiability of F , we know that there exists a sufficiently small $\delta > 0$, such that when $\|y - x^*\| < \delta$,

$$\|F(y) - F(x^*) - F'(x^*)(y - x^*)\| \leq \frac{1}{2\beta}\|y - x^*\| \quad (3.6.20)$$

holds. Note that

$$F(y) = [F'(x^*)(y - x^*)] + [F(y) - F(x^*) - F'(x^*)(y - x^*)],$$

and take norms, then we have

$$\begin{aligned} \|F(y)\| &\leq \|F'(x^*)\|\|y - x^*\| + \|F(y) - F(x^*) - F'(x^*)(y - x^*)\| \\ &\leq \left(\|F'(x^*)\| + \frac{1}{2\beta}\right)\|y - x^*\| \end{aligned} \quad (3.6.21)$$

and

$$\begin{aligned} \|F(y)\| &\geq \|F'(x^*)^{-1}\|^{-1}\|y - x^*\| - \|F(y) - F(x^*) - F'(x^*)(y - x^*)\| \\ &\geq \left(\|F'(x^*)^{-1}\|^{-1} - \frac{1}{2\beta}\right)\|y - x^*\| \\ &= \frac{1}{2\beta}\|y - x^*\|. \end{aligned} \quad (3.6.22)$$

Combining (3.6.21) and (3.6.22) gives (3.6.19). $\square$

Theorem 3.6.4 *Let the assumptions of Theorem 3.6.2 be satisfied. Assume that the sequence $\{x_k\}$ generated by the inexact Newton's method converges to x^* , then, if and only if*

$$\|r_k\| = o(\|F(x_k)\|), \quad k \rightarrow \infty, \quad (3.6.23)$$

$\{x_k\}$ converges to x^* superlinearly.

Proof. Assume that $\{x_k\}$ converges to x^* superlinearly. Since

$$\begin{aligned} r_k &= F(x_k) + F'(x_k)(x_{k+1} - x_k) \\ &= [F(x_k) - F(x^*) - F'(x^*)(x_k - x^*)] - [F'(x_k) - F'(x^*)](x_k - x^*) \\ &\quad + [F'(x^*) + (F'(x_k) - F'(x^*))](x_{k+1} - x^*), \end{aligned}$$

taking norms and using property (A1)-(A3) and the superlinear convergence property of $\{x_k\}$ yield

$$\begin{aligned} \|r_k\| &\leq \|F(x_k) - F(x^*) - F'(x^*)(x_k - x^*)\| + \|F'(x_k) - F'(x^*)\|\|x_k - x^*\| \\ &\quad + [\|F'(x^*)\| + \|F'(x_k) - F'(x^*)\|]\|x_{k+1} - x^*\| \\ &= o(\|x_k - x^*\|) + o(1)\|x_k - x^*\| \\ &\quad + [\|F'(x^*)\| + o(1)]o(\|x_k - x^*\|). \end{aligned} \tag{3.6.24}$$

Thus, by use of Lemma 3.6.3, we have, when $k \rightarrow \infty$, that

$$\|r_k\| = o(\|x_k - x^*\|) = o(\|F(x_k)\|). \tag{3.6.25}$$

Conversely, assume that $\|r_k\| = o(\|F(x_k)\|)$. From (3.6.16), it follows that

$$\begin{aligned} &\|x_{k+1} - x^*\| \\ &\leq (\|F'(x^*)^{-1}\| + \|F'(x_k)^{-1} - F'(x^*)^{-1}\|)(\|r_k\| \\ &\quad + \|F'(x_k) - F'(x^*)\|\|x_k - x^*\| + \|F(x_k) - F(x^*) - F'(x^*)(x_k - x^*)\|) \\ &= (\|F'(x^*)^{-1}\| + o(1))(o(\|F(x_k)\|) + o(1)\|x_k - x^*\| + o(\|x_k - x^*\|)). \end{aligned}$$

Therefore, we get from Lemma 3.6.3 that

$$\begin{aligned} \|x_{k+1} - x^*\| &= o(\|F(x_k)\|) + o(\|x_k - x^*\|) \\ &= o(\|x_k - x^*\|), \end{aligned}$$

which shows the superlinear convergence of sequence $\{x_k\}$. $\square$

The following corollary indicates that when $\{\eta_k\} \rightarrow 0$, the sequence $\{x_k\}$ converges to x^* superlinearly.

Corollary 3.6.5 *Assume that the sequence $\{x_k\}$ generated by inexact Newton's method converges to x^* . Then, if sequence $\{\eta_k\}$ converges to zero, the sequence $\{x_k\}$ converges to x^* superlinearly.*

Proof. If $\lim_{k \rightarrow \infty} \eta_k = 0$, then

$$\limsup_{k \rightarrow \infty} \frac{\|r_k\|}{\|F(x_k)\|} = 0,$$

which means that $\|r_k\| = o(\|F(x_k)\|)$. Then the conclusion is obtained from Theorem 3.6.4. $\square$

There are several proofs of local convergence for the inexact Newton's method. Below, we give outlines of other proofs.

The outline of the second proof is as follows.

From (3.6.4)–(3.6.6), we have

$$\begin{aligned} & x_{k+1} - x^* \\ &= x_k - x^* - F'(x_k)^{-1}F(x_k) + F'(x_k)^{-1}r_k \\ &= F'(x_k)^{-1}[F'(x_k)(x_k - x^*) - F(x_k) + F(x^*) + r_k]. \end{aligned} \quad (3.6.26)$$

Taking norms, and using (3.6.7), (3.6.15) and Lipschitzian continuity of $F(x)$, i.e., $\|F(x_k)\| = \|F(x_k) - F(x^*)\| \leq L\|x_k - x^*\|$, we obtain

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \xi[\gamma\|x_k - x^*\| + \eta_k L\|x_k - x^*\|] \\ &\leq \xi(\gamma + \eta_k L)\|x_k - x^*\|. \end{aligned} \quad (3.6.27)$$

If we choose γ and η_k such that $\xi(\gamma + \eta_k L) < 1$, then $\{x_k\}$ converges to x^* linearly. If we choose $\eta_k \rightarrow 0$ and note that γ is sufficiently small, then $\xi(\gamma + \eta_k L) \rightarrow 0$, and thus the sequence $\{x_k\}$ converges to x^* superlinearly.

The third proof is as follows.

Theorem 3.6.6 *Let $F : R^n \rightarrow R^n$ satisfy the properties (A1)–(A3). Assume that the sequence $\{\eta_k\}$ satisfies $0 \leq \eta_k \leq \eta < 1$. Then, for some $\epsilon > 0$, if the starting point x_0 is sufficiently near x^* , the sequence $\{x_k\}$ generated by inexact Newton's method (3.6.4)–(3.6.6) converges to x^* , and the convergence rate is linear, i.e., for all k sufficiently large,*

$$\|x_{k+1} - x^*\| \leq c\|x_k - x^*\| \quad (3.6.28)$$

for some constant $0 < c < 1$.

Furthermore, if $\eta_k \rightarrow 0$, then the sequence $\{x_k\}$ converges to x^* superlinearly. If $\eta_k = O(\|F(x_k)\|)$, then the sequence converges to x^* quadratically.

Proof. From (3.6.4),

$$s_k = F'(x_k)^{-1}[-F(x_k) + r_k].$$

Taking norms and using (3.6.7) and (3.6.5), we obtain

$$\|s_k\| \leq \xi(\|F(x_k)\| + \|r_k\|) \leq \xi(1 + \eta)\|F(x_k)\| \leq 2\xi\|F(x_k)\|. \quad (3.6.29)$$

By using Taylor's theorem, (3.6.4) and the above expression, we have

$$\begin{aligned} F(x_{k+1}) &= F(x_k) + F'(x_k)s_k + O(\|s_k\|^2) \\ &= r_k + O(\|F(x_k)\|^2). \end{aligned} \quad (3.6.30)$$

By taking norms and using (3.6.5), we get

$$\|F(x_{k+1})\| \leq \eta_k\|F(x_k)\| + O(\|F(x_k)\|^2). \quad (3.6.31)$$

Dividing both sides by $\|F(x_k)\|$, passing to the $\limsup, k \rightarrow \infty$, noting that $\eta_k \leq \eta < 1$, we deduce

$$\limsup_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|F(x_k)\|} \leq \eta < 1. \quad (3.6.32)$$

By using Corollary 1.2.26 (or Lemma 3.6.3) we immediately obtain

$$\limsup_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq C \limsup_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|F(x_k)\|} \quad (3.6.33)$$

for some constant C . When $\{x_k\}$ is sufficiently close to x^* and $C\eta < 1$, the sequence $\{x_k\}$ converges to x^* locally and linearly.

Furthermore, if $\eta_k \rightarrow 0$, then

$$\limsup_{k \rightarrow \infty} \frac{\|r_k\|}{\|F(x_k)\|} = 0,$$

i.e., $\|r_k\| = o(\|F(x_k)\|)$. By using (3.6.30) and taking norms, we have

$$\limsup_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|F(x_k)\|} = 0 \quad (3.6.34)$$

which indicates the superlinear convergence in the function value sequence $\{F(x_k)\}$. It is easy to see that

$$\limsup_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0 \quad (3.6.35)$$

by use of Corollary 1.2.26 (or (3.6.19)).

If

$$\eta_k = O(\|F(x_k)\|), \quad (3.6.36)$$

then there exists some constant c_1 such that $\eta_k \leq c_1\|F(x_k)\|$. By using (3.6.5) we get that

$$\limsup_{k \rightarrow \infty} \frac{\|r_k\|}{\|F(x_k)\|^2} \leq c_1, \quad (3.6.37)$$

which shows that

$$r_k = O(\|F'(x_k)\|^2). \quad (3.6.38)$$

We have immediately from (3.6.30) that

$$\limsup_{k \rightarrow \infty} \frac{\|F(x_{k+1})\|}{\|F(x_k)\|^2} = c \quad (3.6.39)$$

for some constant c , which means quadratic convergence of $\{F(x_k)\}$. And therefore we have that

$$\limsup_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|^2} = c. \quad \square \quad (3.6.40)$$

It is easy to apply the above result to unconstrained optimization problem $\min_{x \in R^n} f(x)$. In fact, instead of (3.6.4)–(3.6.5), we use

$$\nabla^2 f(x_k) s_k = -\nabla f(x_k) + r_k, \quad (3.6.41)$$

where

$$\|r_k\| \leq \eta_k \|\nabla f(x_k)\|, \quad (3.6.42)$$

and then we can get the same results for unconstrained optimization problems. Similar to the above discussion we have the following theorem.

Theorem 3.6.7 *Suppose that $\nabla f(x)$ is continuously differentiable in a neighborhood of a minimizer x^* , and assume that $\nabla^2 f(x^*)$ is positive definite. Consider the iteration $x_{k+1} = x_k + s_k$, where s_k is an inexact Newton step satisfying (3.6.41) and (3.6.42). Assume that the sequence $\{\eta_k\}$ satisfies $0 \leq \eta_k \leq \eta < 1$. Then, if the starting point x_0 is sufficiently near x^* , the sequence $\{x_k\}$ converges to x^* linearly, i.e., for all k sufficiently large,*

$$\|x_{k+1} - x^*\| \leq c \|x_k - x^*\| \quad (3.6.43)$$

for some constant $0 < c < 1$.

The sequence $\{x_k\}$ converges to x^ superlinearly if $\|r_k\| = o(\|\nabla f(x_k)\|)$.*

The sequence $\{x_k\}$ converges to x^ quadratically if $\|r_k\| = O(\|\nabla f(x_k)\|^2)$.*

About the implementation of the inexact Newton's method, we can generate the search direction by applying the conjugate gradient method to the Newton's equation $\nabla^2 f(x_k)s_k = -\nabla f(x_k)$, and then ask that the termination test (3.6.42) be satisfied.

Inexact Newton's method is an efficient method, especially for large scale nonlinear equations and optimization problems. Inexact Newton's method was due to Dembo, Eisenstat and Steihaug [83]. The other important works about this method can be found in Steihaug [321], Dennis and Walker [99], Ypma [365], and Nash [229].

Exercises

1. Let $f(x) = \frac{3}{2}x_1^2 + \frac{1}{2}x_2^2 - x_1x_2 - 2x_1$. Let the initial point $x^{(0)} = (-2, 4)^T$. Minimize $f(x)$ by use of the steepest descent method and Newton's method, respectively.

2. Let

$$(1) f(x) = \frac{1}{2}(x_1^2 + 9x_2^2);$$

$$(2) f(x) = \frac{1}{2}(x_1^2 + 10^4x_2^2).$$

Discuss the convergence rate of the steepest descent method.

3. Let $f(x) = \frac{1}{2}x^T x + \frac{1}{4}\sigma(x^T A x)^2$, where

$$A = \begin{bmatrix} 5 & 1 & 0 & \frac{1}{2} \\ 1 & 4 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 3 & 0 \\ \frac{1}{2} & 0 & 0 & 2 \end{bmatrix}.$$

Let (1) $x^{(0)} = (\cos 70^\circ, \sin 70^\circ, \cos 70^\circ, \sin 70^\circ)^T$;

(2) $x^{(0)} = (\cos 50^\circ, \sin 50^\circ, \cos 50^\circ, \sin 50^\circ)^T$.

In the case of $\sigma = 1$ and $\sigma = 10^4$, discuss the numerical results and behavior of convergence rate of pure Newton's method and Newton's method with line search respectively.

4. Minimize the Rosenbrock function $f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ by the steepest descent method and Newton's method respectively, where $x^{(0)} = (-1.2, 1)^T, x^* = (1, 1)^T, f(x^*) = 0$.

5. By your opinion, state the reasons that the steepest descent method converges slowly.

6. Prove the convergence of the inexact Newton methods.